NIHR | National Institute for Health and Care Research

# Summary

| Reference Number | NIHR305218 |
|---|---|
| Lead Applicant | Miss Sabrena Lee |
| Research Title | Investigating the Nature and Impact of Paediatric Feeding Disorders on Ventilator-Assisted Children and their Families |
| Plain English Summary | **Aims:**<br>1. I will examine how services support ventilator-assisted children with feeding problems. I will collect information from centres in the UK, New Zealand, Australia and Canada.<br>2. I will review patient records at three hospitals in England to:<br>a. identify the number of ventilator-assisted children with feeding problems<br>b. report the type of problems they had<br>3. I will assess ventilator-assisted children and interview their parents/carers three and six months after hospital discharge to:<br>a. investigate the type of feeding problems children have<br>b. explore the psychological and social impacts of children's feeding needs on their families<br><br>**Background:**<br>Ventilator-assisted children need long-term breathing support from a ventilation machine for some or all of the day and night. Children receive ventilation through a face/nasal mask or tracheostomy (a surgically-created opening in the neck). Their medical and ventilation needs may impact their ability to feed in a successful and safe way. In 2019, there were approximately 2383 ventilator-assisted children in the UK. This number is rising around the world. Yet, there are no policies or guidelines about feeding support for these children. There is little research about the number of children with feeding problems and the types of difficulties they have. There is also a large gap in exploring the psychological and social impacts of these issues on families. It is important to investigate these topics so we can understand the needs of these children and their families.<br><br>**Design and methods:**<br>***Project 1:*** *Feeding services in the UK, Canada, Australia and New Zealand*<br>I will collect information from online resources about service policies, guidelines and protocols. I will speak with ventilation centres in these countries and invite health professionals working in hospital and community teams to complete an online survey. This survey will include questions about how health professionals support ventilator-assisted children with face/nasal mask and tracheostomy ventilation who have feeding problems.<br><br>***Project 2:*** *The number of children with feeding problems and the type of problems present*<br>I will review the hospital notes of ventilator-assisted children under 5 years of age with face/nasal mask and tracheostomy ventilation between 2021 to 2024. I will collect information from Evelina London Children's Hospital, Great Ormond Street Hospital and Alder Hey Hospital. I will identify the number of children with feeding problems. I will report on the problems they had at discharge and for 12 months after discharge from hospital.<br><br>***Project 3:*** *The types of feeding problems children have and the impacts on families*<br>I will assess ventilator-assisted children under 5 years of age with face/nasal mask and tracheostomy ventilation who are discharged from hospital needing feeding support. Parents/carers will complete questionnaires about their child's feeding and its impacts. I will assess children's eating and drinking in their home. I will interview parents/carers to explore the psychological and social impacts of their children's feeding on their families. If children have a swallow x-ray study at the hospital, I will re-analyse these recordings.<br><br>**Public involvement and engagement:**<br>Parents of ventilator-assisted children and health professionals have contributed to my |

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|---|---|
| | research proposal and this plain English summary. I will meet with them and other expert health professionals to seek advice during the projects. I will continue working with this group to plan and develop future research work.<br><br>**Sharing research findings:**<br>My public involvement group and I will share research findings to a range of people and forums. We will speak at conferences and meetings, and publish research findings in healthcare journals. I will use social media to provide updates about my research to the public and relevant networks. |

Our purpose for collecting this data is to communicate with you about your application and have the necessary information to evaluate you for a grant. The data we collect here is collected in the public interest. Information provided here may be subject to Freedom of Information requests.

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The NIHR privacy policy includes further information including ways we may use your data, our contact details and details on your individual rights regarding how your data is used. Your data may be shared across the NIHR, including with other coordinating centres, to allow the application to be managed and for statistical analysis, and with external grant reviewers as part of the process for managing the allocation of a grant. This notice is under constant review and will be updated and / or revised based on that review as appropriate.

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## 1. Application Summary Information

| **Host organisation (which will administer any award)** |
|---|
| Guy's and St Thomas' NHS Foundation Trust |

| **Partner organisation** | King's College London |
|---|---|

| **Research Title** | Investigating the Nature and Impact of Paediatric Feeding Disorders on Ventilator-Assisted Children and their Families |
|---|---|

| **Partnership Fellowship**<br>Please select the relevant Partnership Fellowship. |
|---|
| N/A |

| **Proposed WTE**<br>Please select which WTE option you would prefer |
|---|
| 1.00 |

| **Duration (months)**<br>The duration for your award is based on your chosen WTE |
|---|
| 36 |

| **Proposed start date if grant awarded** | 01/09/2025 |
|---|---|

| **Research Type** | Qualitative study, Retrospective review, Evidence Synthesis |
|---|---|

## 2. Applicant CV

A CV for the Applicant is a mandatory component of this application; please ensure this is completed fully.
*Please note that some of the CV questions are automatically completed from information you have provided in the 'Update CV' section of the 'Manage My Details' page. These can be updated via the ARAMIS Home Page or by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx*

**Degrees and Professional Qualifications**

| From | To | Qualification | Subject | Country | Institution | Class |
|---|---|---|---|---|---|---|
| 09/2021 | 08/2022 | MRes | Master of Research in Speech, Language and Cognition | United Kingdom (England) | University College London | Distinction |
| 02/2007 | 12/2010 | BSc(Hons) | Bachelor of Speech Pathology | Australia | University of Queensland | First Class Honours |

**Present and previous positions**
(including from/to dates, job title and employer)

| From | To | Position | Organisation | WTE % Research |
|---|---|---|---|---|
| 09/2018 | - | Highly Specialist Speech and Language Therapist | Guy's and St Thomas' NHS Foundation Trust | 15 |
| 06/2017 | 06/2018 | Speech Pathologist | The Children's Hospital at Westmead | 15 |
| 02/2016 | 07/2017 | Speech Pathologist | Liverpool Hospital | 15 |
| 03/2015 | 01/2016 | Speech Pathologist | Lady Cilento Children's Hospital | 15 |
| 01/2015 | 03/2015 | Speech Pathologist | SDN Children's Services | 15 |
| 04/2013 | 12/2014 | Speech Pathologist | The Children's Hospital at Westmead | 5 |
| 03/2013 | 12/2014 | Research Assistant | The Children's Hospital at Westmead | 50 |
| 05/2012 | 04/2013 | Speech Pathologist | St George and Caringbah Hospitals | 15 |
| 01/2011 | 05/2012 | Speech Pathologist | Education Queensland | 15 |

**Research Grants held**

| Title of Grant | Role in the Research Grant | Total Grant £ | Source of Grant (Organisation Name) | Duration (In Months) | Start Date of Funding | End Date of Funding |
|---|---|---|---|---|---|---|
| Research Design Service Public Involvement Fund | This was a pre-award offering for researchers submitting an application to any peer-reviewed applied health and social care research funding streams in England. I was awarded this funding in September 2023 and used it to conduct patient and public involvement activity to prepare this fellowship application. Application ID: e598. | 210 | Research Design Service | 9 | 20/09/2023 | 11/06/2024 |
| HEE/NIHR ICA Pre-Doctoral Clinical Academic | I completed this research fellowship in 2022 (grant number: NIHR301904) and undertook a Master of Research degree at University College London. Please see 'please describe your research career to date' section for details on my roles in this fellowship. Main related publication | 72979 | HEE/NIHR | 12 | 01/09/2021 | 01/09/2022 |

| Title of Grant | Role in the Research Grant | Total Grant £ | Source of Grant (Organisation Name) | Duration (In Months) | Start Date of Funding | End Date of Funding |
|---|---|---|---|---|---|---|
| Fellowship | DOIs: 1) 10.1177/13674935241242824, 2) 10.1371/journal.pone.0287872 and 3) 10.1007/s00455-023-10648-0. | | | | | |

**Publication Record**

This should display the authors, date, article title and journal name for each publication added. This information can be edited via the "My Research Outputs" page of your ARAMIS account.

| Publications |
|---|
| Lee, Marshall, Clarke and Smith. (2024) ''They don't know what to do with our children': Experiences and views on feeding and swallowing from parents of children who use long-term ventilation', *Journal of Child Health Care*, (Epub ahead of print). |
| Lee, Marshall, Clarke and Smith. (2024) 'Feeding and swallowing outcomes of children receiving long-term ventilation: A scoping review protocol', *PloS one*, 2(19). |
| Lee, Marshall, Clarke and Smith. (2024) 'Feeding and Swallowing Outcomes in Children Who Use Long-Term Ventilation: A Scoping Review', *Dysphagia*, (Online ahead of print). |
| Sutherland, R., Trembath, D., Hodge, A., Drevensek, S., Lee, S., Silove, N. and Roberts, J.. (2016) 'Telehealth language assessments using consumer grade equipment in rural and urban settings: Feasible, reliable and well tolerated.', *Journal of Telemedicine and Telecare*, 23(1). |
| Hemsley, B., Lee, S., Munro, K., Seedat, N., Bastock, K. and Davidson, B.. (2013) 'Supporting communication for children with cerebral palsy in hospital: Views of community and hospital staff.', *Developmental neurorehabilitation*, 17(3). |

**Relevant Prizes, Awards and other Academic Distinctions**

Please note the information for additional prizes or awards is automatically populated from information provided in the "Update CV" section of the "Manage My Details" page, which you can update by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

| Date (mm/yyyy) | Source | Prize | Role of Applicant |
|---|---|---|---|
| 01/12/2022 | University College London | Dean's List for Master of Research in Speech, Language and Cognition | Student |

**ORCID iD** 0000-0002-0653-448X

## 3. Applicant Research Background

| **Professional Background** | Allied Health Professional |
|---|---|

| **Profession (Allied Health Professional)** | Speech and language therapist |
|---|---|

| **Do you have current / active professional registration?**<br>This registration must be with one of the **ICA approved regulatory bodies** |
|---|
| Yes |

| **Professional registration number/pin**<br>Please supply a valid professional registration number/pin followed by the regulatory body (eg 12A3456B - NMC or PA12345 - HCPC) |
|---|
| SL35968 |

**Please describe your research career to date**

**Research Experience**

Health Education England/National Institute for Health and Care Research Pre-doctoral Clinical Academic Fellowship:

I completed a Health Education England (HEE)/National Institute for Health and Care Research (NIHR) Pre-doctoral Clinical Academic Fellowship (PCAF) in 2022 and undertook a Master of Research (MRes) in Speech, Language, and Cognition at University College London (UCL). I was awarded a distinction and named on the Dean's List for outstanding academic achievement. During this fellowship, I completed a scoping review and a qualitative research activity.

*Scoping Review*

This scoping review investigated the feeding and swallowing outcomes of ventilator-assisted children. I was responsible for formulating the research question, planning the protocol, designing and implementing a search strategy, compiling and screening literature, extracting data and synthesising findings. I gained valuable skills in reviewing and appraising the evidence-base, and have a thorough understanding of available evidence and gaps within my research topic. I am lead author on two resulting publications: a scoping review protocol and a scoping review[1,2].

*Qualitative Research*

This research activity investigated the perspectives and experiences of feeding and swallowing of parents of ventilator-assisted children. This was conducted in collaboration with WellChild charity. My roles included formulating the research question, designing methodology, obtaining ethical and data protection approval, conducting and transcribing interviews, and analysing data. I am lead author on a publication resulting from this work[3].

*Research dissemination and impact*

I have disseminated my PCAF findings across national and international forums. I have established collaborative links with clinical and research teams at Evelina London Children's Hospital, Pan-Thames Paediatric Long-Term

Ventilation Network and WellChild charity. I presented to the Council for Allied Health Professions Research (50 audience members) and the National Allied Health Professionals Long-Term Ventilation network (60 audience members) in the United Kingdom (UK). I disseminated findings to the Paediatric Dysphagia Interest Group (over 90 participants) in Australia. My research was featured in a blog post in the Feeding Matters conference which was held virtually from the United States and streamed worldwide.

Other methodologies and publications:

In 2010, I conducted a qualitative study during my Honours degree exploring the views of health professionals on communication support for children with cerebral palsy. In 2014, alongside my clinical role, I was a research assistant at Westmead Children's Hospital in Australia. I worked on a study investigating the applicability of telehealth in speech and language therapy assessments. I used qualitative and quantitative methods to collect survey data, conduct patient assessments, design and complete inter-rater reliability tools, and disseminate findings. I co-authored on publications during my Honours degree in 2014 and research assistant work in 2016[4,5].

Research training:

In 2020, I completed the Designing Clinical Research course through King's Health Partners and University of California San Francisco. This three-month course provided training on designing research questions, collating background information, planning subjects/sample size, study methodology, results analysis, statistics and ethics. I designed, wrote and presented a complete research proposal.

During my MRes, I gained foundational skills in professional research practice which provided me with essential preparation for undertaking a Doctor of Philosophy (PhD). During this time, I attended external training courses including the UCL Hospital Biomedical Sciences Patient and Public (PPI) Involvement workshop and the UCL Academic Communication Centre writing course. I have completed peer reviewer training and International Council on Harmonisation of Technical Requirements good clinical practice training.

Research networks:

I am on the committee for the National Allied Health Professionals Long-Term Ventilation Group. This involves regular coordination of professional development opportunities for network members. I have presented education and training to this network on several occasions. I am a member of the Paediatric Dysphagia Speech and Language Therapists in Research network. My membership in NIHR Ear, Nose and Throat Speech and Language Therapy Champions provides me with support from NIHR research leaders. Mentoring is provided through my hospital trust's doctoral peer support group. I am a reviewer for the NIHR Evidence Alerts database and actively peer-review for Joanna Briggs Institute Evidence Synthesis.

**Professional Experience**

I have been a speech and language therapist for 14 years and am the respiratory lead in my team at Evelina London Children's Hospital. This involves providing services to children with a variety of feeding and communication difficulties. I specialise in respiratory diseases, tracheostomy and ventilation. I enjoy working with a variety of multidisciplinary teams and look forward to extending my scope of practice to become a clinical academic leader in this area.

Over my career, I have taken part in supervision, clinical education, service evaluation and department-based research projects in keeping with the National Health Service clinical governance framework. I have used my research skills to guide evidence-based practice, clinical decision making and professional development. I have led quality improvement projects including the development of an inpatient long-term ventilation service and adaptation of this service during the Coronavirus disease pandemic.

**Career trajectory:**

During the PCAF, I thoroughly enjoyed working collaboratively with families and stakeholders to address research priorities. I feel compelled to continue advocating for ventilator-assisted children and their families. I have continued to engage with parents involved in my qualitative study who have since formed part of my PPI group. Their engagement and contributions have been valuable in the preparation of this research proposal and fellowship application. My publications have inspired discussion in national and international teams who have invited me to present about my research outcomes (see 'research dissemination and impact' above). I have also provided support and mentoring to other health professionals who are seeking clinical academic careers.

During the PhD, I will explore current research gaps in the topic of feeding and swallowing in ventilator-assisted children. I am determined to drive robust evidence-based initiatives in the UK and abroad, and will continue to develop local and international research networks. I am committed to expanding my clinical and leadership skills in order to become an expert practitioner academic. I am excited to continue my clinical academic career in which my specialist practice is intertwined with a lifelong commitment to research.

| **Have you already registered for a PhD?** | No |
|---|---|

| **Has this application, or a similar application, previously been submitted to this or any other funding body?** |
|---|
| No |

**Contextual Factors**
Please use this question to detail any contextual factors you wish to make the Selection Committee aware of. NIHR wants to know about any circumstances so that they may take them into consideration during the assessment of your application. Contextual factors may include:

- Career breaks due to parental leave, or periods of illness.
- Reduced time spent undertaking research due to a disability or caring responsibilities. This could include any physical or mental difficulty that may have impacted your research career. These are situations that have a significant impact on your ability to undertake research.
- Reduced opportunities to career support e.g. mentorship, and limited opportunities to undertake prior research and training.
- Impact of the COVID-19 pandemic on your research career.

Please also use this section to detail any other factors that may have impacted your research career not listed in the examples provided. The impact on your career to date will be specific to your particular circumstances but could include such impacts as limited opportunities to obtain grant funding or fewer publications. In general terms, mitigating factors should be significant, and relevant.

NIHR acknowledges that you may be reluctant, or uncomfortable disclosing relevant information that is sensitive. However, you should bear in mind that we are unable to take into account factors that you do not disclose. Please be assured that the information provided by you is sensitive and will be treated confidentially and in line with General Data and Protection Regulations (GDPR).

I took 12 months of maternity leave between December 2022 to December 2023. During this time, I used my Keeping In Touch days to maintain my clinical skills in specialist areas, keep up to date with mandatory training, and train in electronic systems introduced to our hospital. I used my personal time to maintain my research knowledge and skills in order to prepare this Doctoral Clinical and Practitioner Academic Fellowship (DCAF) application. I formed my supervisory team and met with them regularly throughout my maternity leave to prepare my DCAF application. I have continued to engage with my PPI group and was awarded a Public Involvement Fund grant for this PPI work (see 'Research Grants held' in Applicant CV section). I continued to attend research network meetings and published my PCAF research outcomes in three separate manuscripts. I have continued to develop my research and clinical networks, and have presented about the topic of feeding and swallowing in long-

term ventilation to various multidisciplinary teams across national and international forums. Despite being on maternity leave, I have worked hard to develop my specialist clinical and research skills, and continue my commitment to a career as a clinical academic.

## 4. Plain English Summary of Research

**Aims:**

1. I will examine how services support ventilator-assisted children with feeding problems. I will collect information from centres in the UK, New Zealand, Australia and Canada.
2. I will review patient records at three hospitals in England to:
    a. identify the number of ventilator-assisted children with feeding problems
    b. report the type of problems they had
3. I will assess ventilator-assisted children and interview their parents/carers three and six months after hospital discharge to:
    a. investigate the type of feeding problems children have
    b. explore the psychological and social impacts of children's feeding needs on their families

**Background:**

Ventilator-assisted children need long-term breathing support from a ventilation machine for some or all of the day and night. Children receive ventilation through a face/nasal mask or tracheostomy (a surgically-created opening in the neck). Their medical and ventilation needs may impact their ability to feed in a successful and safe way. In 2019, there were approximately 2383 ventilator-assisted children in the UK. This number is rising around the world. Yet, there are no policies or guidelines about feeding support for these children. There is little research about the number of children with feeding problems and the types of difficulties they have. There is also a large gap in exploring the psychological and social impacts of these issues on families. It is important to investigate these topics so we can understand the needs of these children and their families.

**Design and methods:**

***Project 1:*** *Feeding services in the UK, Canada, Australia and New Zealand*

I will collect information from online resources about service policies, guidelines and protocols. I will speak with ventilation centres in these countries and invite health professionals working in hospital and community teams to complete an online survey. This survey will include questions about how health professionals support ventilator-assisted children with face/nasal mask and tracheostomy ventilation who have feeding problems.

***Project 2:*** *The number of children with feeding problems and the type of problems present*

I will review the hospital notes of ventilator-assisted children under 5 years of age with face/nasal mask and tracheostomy ventilation between 2021 to 2024. I will collect information from Evelina London Children's Hospital, Great Ormond Street Hospital and Alder Hey Hospital. I will identify the number of children with feeding problems. I will report on the problems they had at discharge and for 12 months after discharge from hospital.

***Project 3:*** *The types of feeding problems children have and the impacts on families*

I will assess ventilator-assisted children under 5 years of age with face/nasal mask and tracheostomy ventilation who are discharged from hospital needing feeding support. Parents/carers will complete questionnaires about their child's feeding and its impacts. I will assess children's eating and drinking in their home. I will interview parents/carers to explore the psychological and social impacts of their children's feeding on their families. If children have a swallow x-ray study at the hospital, I will re-analyse these recordings.

**Public involvement and engagement:**

Parents of ventilator-assisted children and health professionals have contributed to my research proposal and this plain English summary. I will meet with them and other expert health professionals to seek advice during the projects. I will continue working with this group to plan and develop future research work.

**Sharing research findings:**

My public involvement group and I will share research findings to a range of people and forums. We will speak at conferences and meetings, and publish research findings in healthcare journals. I will use social media to provide updates about my research to the public and relevant networks.

## 5. Scientific Abstract

Background: Ventilator-assisted children have underlying medical conditions that place them at risk of feeding difficulties. In 2019, there were approximately 2383 ventilator-assisted children in the UK. This number is anticipated to rise due to ongoing advancements in medical technology. However, there are no national policies or guidelines about feeding support for this population. There is limited research about the prevalence and types of feeding difficulties children present with. Moreso, the psychosocial impact of feeding difficulties on families has not been thoroughly explored. Therefore, we have little understanding about the experiences and challenges faced by children and their families. Their contributions are key in identifying future research priorities and addressing their health and social care needs.

Overall research aim: To explore current practices/policies and investigate the occurrence, nature and psychosocial impact of paediatric feeding disorders (PFDs) on ventilator-assisted children and their families.

Research objectives:

I will complete three studies investigating the needs of children receiving invasive and non-invasive long-term ventilation:

1. Examine current national and international feeding service provision, policies and clinical practices.
2. Retrospectively identify the proportion of children under 5 years of age who presented with PFD at three hospitals, and characterise the nature of their PFDs during and up to 12 months following hospital discharge.
3. Prospectively assess and characterise the nature of PFDs of children under 5 years of age, and explore the psychosocial impact of PFD on their families 3 and 6 months following hospital discharge.

Methods:

1) Environmental scan: I will synthesise evidence from online databases and grey literature about feeding services, policies and protocols for ventilator-assisted children. I will gather information from stakeholders across the UK, Canada, Australia and New Zealand. An online survey of health professionals in these countries will also be conducted.

2) Retrospective cohort study: I will review the hospital records of ventilator-assisted children discharged from Great Ormond Street Hospital, Evelina London Children's Hospital and Alder Hey Hospital between 2021 to 2024. I will identify the proportion of ventilator-assisted children with PFD and characterise feeding difficulties present at discharge and up to 12 months following discharge.

3) Prospective case-series study: I will recruit ventilator-assisted children who are discharged from Great Ormond Street Hospital and Evelina London Children's Hospital with a referral for community feeding support. I will characterise children's PFD using parent questionnaires, video assessment and secondary analysis of instrumental assessment. I will explore psychosocial impacts of feeding difficulties on families during interviews with parents/caregivers.

Timelines for delivery:

- March 2026: Research regulatory processes
- August 2026: Environmental scan
- May 2027: Retrospective cohort study
- April 2028: Prospective case-series study
- September 2028: PhD Thesis

Anticipated impact and dissemination:

This fellowship will meet research gaps in the fields of paediatric feeding and long-term ventilation. I will disseminate findings through networks, relevant conferences and peer-reviewed journals. Research outcomes will inform planning of research priorities for my post-doctoral work, including the development of evidence-based practice guidelines and interventions to improve the feeding outcomes of children with ventilation needs.

## 6. Detailed Research Plan

**1. What is the problem being addressed?**

Introduction:

Ventilator-assisted children require long-term support from a ventilator to sustain or optimise their breathing[6]. Ventilation is delivered non-invasively (via a face/nasal mask) or invasively (via a surgically-created opening in the neck called a tracheostomy) for part or all of the day and night[6]. Ventilator-assisted children are a vulnerable group of patients who have multiple long-term conditions/multimorbidity including respiratory difficulties, musculoskeletal issues and/or central nervous system impairments[7,8]. In 2019, there were an estimated 2383 ventilator-assisted children in the UK[9]. This number is anticipated to increase in the UK and globally due to advancements in medical technology and earlier survival of premature babies[6,9,10].

Children with paediatric feeding disorders (PFDs) present with impaired oral intake that is not typical for their age and is associated with medical, nutritional, feeding skill, and/or psychosocial dysfunction[11]. These multi-faceted problems significantly impact on daily life, restrict children from achieving age-appropriate milestones and limit participation in normal childhood experiences[11]. There are significant research gaps about a) PFDs and b) the psychosocial impacts of PFDs on ventilator-assisted children and their families.

Problem 1: Research about paediatric feeding disorders in ventilator-assisted children is limited

A small body of research has found that ventilator-assisted children have poor feeding and swallowing outcomes in the short and long term[1,12–15]. This research suggests that children are at high risk of dysphagia (swallowing difficulty) and cannot swallow particular food and fluid textures safely[1]. They are often reliant on nasogastric or gastrostomy feeding tubes to receive nutrition for some or all of their diet[1]. However, there are large research gaps about the prevalence and nature of PFD in this population. Further research is required to understand their feeding journey following initiation of ventilation and once discharged home from hospital. Investigating these issues is key to understanding the true occurrence and impact of PFDs in ventilator-assisted children, and how best to devise interventions to address these issues in the future.

Problem 2: There is limited research on the psychosocial impacts of paediatric feeding disorders on ventilator-assisted children and their families

PFDs impact many aspects of life for parents of children without ventilation, including higher anxiety, increased stress and lower self-esteem[16]. Parents report that gastrostomy feeding is demanding, time-consuming and limits their participation in everyday activities[17]. Parents in my qualitative research study discussed impacts on family bonding, social participation, time and family resources[3]. Ventilator-assisted families report a lower quality of life than children with other chronic illnesses and the general population[18]. However, little research currently exists on the psychosocial impacts of PFDs on ventilator-assisted children and their families. This is a significant research gap that must be explored before we can identify ways to improve children's feeding journeys.

**2. Why is this research important?**

Increasing knowledge and understanding about paediatric feeding disorders in children with complex health needs is a current priority in the UK

A priority for global health research is to improve the lives of people with multiple long-term conditions/multi-morbidity through research[8,19]. In 2022, The Royal College of Speech and Language Therapists identified the top ten research priorities in dysphagia[20]. These included "what are the reported psychosocial effects of nasogastric and gastrostomy feeding?" and "is feeding via gastrostomy effective in improving health and wellbeing outcomes of children with neurodisability and dysphagia and their parents?"[20]. I will address these priorities in my research proposal by exploring the psychosocial impact of PFDs on this population.

The feeding and swallowing problems of ventilator-assisted children place a significant economic burden on the health system

Ventilator-assisted children have a significantly higher mortality, increased emergency department presentations and longer hospital admissions than children with other chronic conditions[21]. PFDs can lead to adverse outcomes including aspiration pneumonia, malnutrition, failure to thrive and dehydration[11]. The cost of staying on a long-term ventilation and intensive care unit is £301,888 and £630,388 per year respectively[22]. The average cost of maintaining a gastrostomy tube is £700 per year[23]. Appropriate management of PFDs is crucial to preventing unnecessary economic burdens on the National Health Service. By examining current service delivery and clinical practices, as well as the frequency and severity of PFD, we will understand the true impact of feeding problems on our health and social care systems.

**3. Review of existing evidence**

National and international services, policies and practices for ventilator-assisted children with paediatric feeding disorders

Ventilator-assisted children receive care from large multidisciplinary teams[24]. In the UK, speech and language therapists conduct assessments and treatments to improve safe feeding and swallowing skills[25]. Speech and language therapists contribute vital information to the multidisciplinary team to allow for holistic patient-centred decision-making[24]. However, there are no national clinical practice guidelines, service policies or protocols regarding support for ventilator-assisted children with PFD. Therefore, we have limited knowledge about evidence-based service delivery, centre policies and care practices. We also have little understanding about the feeding and swallowing services that international centres provide. This data is required to inform the development of interventions to improve the feeding journeys of these children and their families.

The nature of paediatric feeding disorders in ventilator-assisted children

Children who require short-term ventilation (for less than three months) experience dysphagia due to their respiratory and co-morbid medical needs[15]. The presence of ventilation support and feeding tubes around the mouth and face result in negative experiences that lead to chronic feeding difficulties such as oral aversion and learnt refusal behaviours[26–28]. Research indicates that longer periods of ventilation result in worse feeding and swallowing outcomes[29].

My scoping review investigated the feeding and swallowing outcomes of ventilator-assisted children[1]. 31 studies were included in the scoping review. Only 5 studies contained more than 100 participants, 5 studies contained 50-100 participants, and 21 studies contained less than 50 participants. Most studies investigated medical aspects of ventilation (such as treatment outcomes and survival rates) and only 23% of studies focused on feeding outcomes (feeding characteristics, tube feeding, assessment and intervention). Studies identified saliva management issues, difficulty eating solids and drinking liquids, oral aversion and oral hypersensitivity. However, this research was limited and non-specific in its reference to PFD. Outcomes were not described using consistent terminology and were not characterised using the formal PFD framework. Reporting of assessment findings was not detailed e.g., did not describe the volumes and textures of food/drink tested, which affected the ability to draw evidence-based conclusions. Therefore, research gaps continue to exist about the prevalence, severity and characteristics of PFD experienced by this population.

The psychosocial impacts of paediatric feeding disorders on ventilator-assisted children and their families

Research about children without ventilation needs has shown that PFDs negatively affect the wellbeing and quality of life of children and their families[30,31]. In my qualitative research study, parents reported similar impacts on children, their siblings and other family members[3]. These discussions were insightful and revealed the potential psychosocial issues that many ventilator-assisted children and their families experience. However, this study contained a small self-selected sample of seven mothers. We need to conduct more in-depth research with a

larger sample of parents/caregivers from a diverse range of ethnic, gender, geographical and socioeconomic backgrounds. This will ensure more accurate representation of the patient population and facilitate a holistic understanding of the impact of PFD in this population.

**4. Research Aims and Objectives**

Overall research aim: To explore current practices and policies, as well as investigate the occurrence, nature and psychosocial impact of PFDs on ventilator-assisted children and their families.

Research objectives:

1. Work package 1: To examine current national and international service provision, policies and clinical practices used to support ventilator-assisted children with PFD.

2. Work package 2: To retrospectively identify the proportion of ventilator-assisted children under 5 years of age who presented with PFD at three different children's hospitals, and to characterise the nature of their PFDs (across medical, nutritional, feeding skill and psychosocial domains) during and up to 12 months following hospital discharge.

3. Work package 3: To prospectively assess and characterise the nature of PFDs (across medical, nutritional, feeding skill and psychosocial domains) in ventilator-assisted children under 5 years of age, and explore the psychosocial impact of PFD on their families 3 and 6 months following hospital discharge.

**5. Project Plan**

I will conduct a series of studies employing quantitative and qualitative methodologies. I will form an expert advisory group of health professionals and parents/caregivers to guide research planning, implementation and dissemination. The outcomes of these projects will inform the development of tools to support families and healthcare professionals in the management of these children in my post-doctoral work.

**Work package 1: *Environmental scan***

I will complete an environmental scan examining current services, policies and practices that multidisciplinary ventilation teams use to support PFDs in the UK, Australia, New Zealand and Canada. I have selected these countries as they have similar healthcare systems to the UK (i.e., healthcare is publicly funded). This environmental scan can inform future planning and implementation of national service delivery, policies and clinical practices.

Design:

I will use the seven-step environmental scan approach proposed and used by Wilburn and colleagues, as it is applicable to healthcare and the objectives of this work package[32]:

1. *Determine leadership and capacity for the project:* Together with my expert advisory group, I will lead the process from development to dissemination under the guidance of my supervision team.

2. *Establish the focal area and purpose:* The purpose is to collect and synthesise information on services, policies and practices about feeding and swallowing management for ventilator-assisted children.

3. *Create and adhere to a timeline and set incremental goals:* I provide an overview of the timeline in my Gantt chart and this will be further detailed during planning with my advisory group.

4. *Determine information to be collected and 5. Identify and engage stakeholders:*
   1) I will conduct a review of peer-reviewed and grey literature including evidence reviews, guidelines, policy documents, websites, letters/opinion pieces and research studies. I will search online databases (PubMed, MEDLINE (Ovid Version), Cochrane library, Web of Science and CINAHL Plus) using a search strategy created with a research librarian. Studies published from 1980 will be included as the research

field of long-term ventilation has significantly developed since this time. I will screen studies according to the eligibility criteria below. Relevant citations will be uploaded into Covidence and extracted using a bespoke extraction tool. I will search for grey literature using handsearching via a generic search engine in Google.

Eligibility criteria:

- Population: children aged 0-5 years
- Subject: long-term ventilation (invasive and non-invasive ventilation provided for at least 3 months from time of initiation and once medically stable)
- Context: feeding and swallowing

2) I will seek relevant stakeholders (e.g. respiratory teams at Great Ormond Street Hospital in the UK, The Hospital for Sick Children in Canada, Sydney Children's Hospital in Australia and the Starship Hospital in New Zealand) through professional societies, my fellowship and health networks, and professional networks of my supervisors/expert advisory panel. I will communicate with a key contact from each stakeholder group through email. I anticipate a sample of 20-30 stakeholder groups based on engagement with at least 5 service providers in each country. I will obtain information/documentation about service delivery, team members, policies, protocols and practices. Information will be entered into the bespoke extraction tool used in step 1.

3: I will design and distribute an online survey. The online survey will examine clinical practice, assessment and intervention tools, barriers and facilitators to service provision, and research priorities. Survey development will be informed by literature review and stakeholder engagement processes. Previous/existing surveys in this topic will also inform survey design. I will meet with my advisory group and supervisory team to reach consensus about final item selection. The survey will be pilot tested for content validity using purposive sampling of 5 hospital sites in the UK. The survey will be conducted on the Qualtrics website.

Health professionals in acute and community settings in all four countries will be invited to participate. Ventilations teams typically consist of a respiratory consultant, specialist nurse, physiotherapist, occupational therapist, psychologist, speech and language therapist and dietician[6]. I anticipate participation from at least 5 ventilation teams in each country. If international centres have similar teams as the UK, I predict a sample of 100 participants.

6. *Analyse and synthesise results into summary report:*
   I will analyse data using Statistical Package for the Social Sciences (SPSS) and triangulate data according to the environmental scan. I will conduct a gap analysis to identify missing information.

   Online survey responses will be analysed using SPSS software. I will use descriptive statistics (counts and percentages) to analyse quantitative findings. I will conduct summative content analysis of qualitative responses to open-ended questions. I will code qualitative responses, check codes with the supervisory team then quantify codes across the data. The Checklist for Reporting Results of Internet E-Surveys will be used to report survey data[33].

7. *Disseminate results and conclusions to key stakeholders:* See 'dissemination, outputs and anticipated impacts' below.

**Work package 2: *Retrospective cohort study***

Design:

I will conduct a retrospective observational cohort study using data from medical records of ventilator-assisted

patients from Great Ormond Street Hospital, Evelina London Children's Hospital and Alder Hey Hospital between the years of 2021 to 2024. A retrospective study will allow me to obtain data about the occurrence of feeding problems (in these hospitals) and report on the types of problems experienced. These hospitals are specialist tertiary centres with large catchment areas across London, and northwest and southeast England. Participants will therefore represent a diverse range of demographic, social and ethnic backgrounds.

I will collect data from 2021 given the integrity of electronic medical records from this date onwards. I will review inpatient and outpatient medical records, including those documented at time of discharge and for 12 months following discharge.

I will include data from children aged 0 to 5 years as this is a critical period of development for feeding skills[34]. There are approximately 20 children initiated on long-term ventilation (aged 0–5 years) at each site per year (as per hospital databases). Therefore, I anticipate a sample of approximately 180 children.

Inclusion criteria:

- Children aged 0–5 years discharged following initiation of invasive or non-invasive long-term ventilation between 2021 to 2024

Exclusion criteria:

- Children in whom ventilation was not long-term (discontinued within three months from initiation and once medically stable)

Data collection and analysis:

1. I will screen patient records from September 2021 to September 2024 and collect data on demographics and medical history.

2. I will identify the proportion of ventilator-assisted patients with PFD using diagnostic criteria proposed by Goday and colleagues: disturbance in oral intake of nutrients, inappropriate for age, lasting at least 2 weeks and associated with one or more of the following: medical, nutritional, feeding skill and/or psychosocial dysfunction[11].

3. I will extract data on the characteristics of PFD at hospital discharge and for 12 months after discharge to populate the following:

- *PFD Case Report Form (CRF)*[35] – this form characterises PFD across medical, nutritional, feeding skill and psychosocial domains.
- *Rosenbek Penetration-Aspiration Scale (PAS)*[36] – a rank ordered scale from 1 to 8 that classifies depth of food/drink entering the airway in videofluoroscopic swallow studies. I will score the PAS based on videofluoroscopic swallow study reports (if information is available).
- *Consensus on Level Descriptors for a Functional Children's Eating and Drinking Activity Scale (CEDAS)*[37] – a scale from 1 to 6 that describes a child's oral/non-oral feeding status. I will report the score on discharge and 12 months after discharge.

4. I will record and code data in a password-protected Excel spreadsheet then transfer it into SPSS for analyses. All demographic, medical history and assessment information will be presented using descriptive statistics. Categorical data will be presented as counts and percentages. Continuous data will be presented as means and standard deviations or medians and interquartile ranges depending on data distribution.

**Work package 3: *Prospective case series study***

I will conduct a prospective mixed-methods case series study. This will allow me to collect data that will inform my research objectives with a feasible sample of patients and their families within the fellowship timeframes.

I will invite patients and their parents/caregivers at Evelina London Children's Hospital and Great Ormond Street Hospital who are being discharged home with a referral for ongoing feeding support. I will recruit families between March 2026 and March 2027.

Inclusion criteria:

- Children aged 0 – 5 years being discharged following initiation of invasive or non-invasive long-term ventilation with a referral to a community speech and language therapist for feeding support
- Families who pass the screening risk assessment for a home visit (e.g., no active safeguarding investigation, no known domestic violence situation)
- Families living in England

Approximately 15 ventilator-assisted children who meet above inclusion criteria are discharged from each site per year (as per hospital databases). Allowing for recruitment of 60-70% of eligible patients, I anticipate a sample of 18-21 children and their parents/caregivers.

Data collection:

Children will be recruited prior to discharge from hospital and followed up at 3 and 6 months after discharge. I will complete the following steps:

1. I will collaborate with respiratory and speech and language therapy teams to screen children who meet the inclusion criteria. The clinical team will inform parents/caregivers about the study. If they express interest, I will meet with them in the hospital to explain the study and provide information sheets/consent forms. Parents/caregivers will be given at least 24 hours to consent and return forms. Translation of research participant material into other languages will be completed as needed.

2. I will access participants' medical records and collect data about demographics and medical history.

3. I will schedule three sessions with the family:

a. 3 months post-discharge: an online one-to-one interview with parents/caregivers
b. 6 months post-discharge: a home visit
c. 2 weeks after the home visit: an online one-to-one interview with parents/caregivers

4. Three months after discharge, I will conduct qualitative parent/caregiver interviews over Zoom/Teams/phone call (depending on their preference) to allow for convenience and flexibility as per families' needs. I will use a topic guide informed by the evidence base and developed with my expert advisory group. This will include questions about the impact of the child's PFD on their family, daily activities, emotional functioning, social functioning and quality of life. Open-ended questions will be used to encourage participant-led discussion. I anticipate interviews to run for 60 minutes. Online interviews will be recorded and transcribed using Zoom/Teams software. Phone calls will be recorded and transcribed by myself. Interpreting services will be used.

5. One week prior to the home visit, I will send families two parent questionnaires to complete and return by email. If families don't have email, they will complete questionnaires during the home visit.

- *The Paediatric Eating Assessment Tool (Pedi-EAT)*[38] – a questionnaire used to assess characteristics of PFD. Parent/caregivers rate the frequency of physiological symptoms, problematic mealtime behaviours, selective/restrictive eating and oral processing difficulties.
- *The Feeding/Swallowing Impact Survey (FS-IS)*[39] – a questionnaire used to measure psychosocial impacts of children's feeding on their parent/caregiver. Parent/caregivers rate the frequency of impacts on daily activities, worrying and feeding.

6. During the home visit, I will conduct an assessment of the child's feeding and swallowing skills by recording the child eating/drinking during a mealtime.

After each home visit, I will analyse the video and characterise children's PFD using:

- *PFD CRF*[35] – described in work package 2
- *CEDAS*[37] – described in work package 2

7. I will conduct a second qualitative interview with each parent/caregiver at 6 months post-discharge. This interview will use the same process as step 4 and aim to obtain a review of their views and experiences on the same questions and topics.

8. If a child receives a videofluoroscopic swallow study during the study period as part of the management plan prescribed by their community speech and language therapist, I will conduct secondary analysis of study recordings. A videofluoroscopic swallow study is an instrumental assessment conducted by speech and language therapists and radiographers in an x-ray room. Children eat/drink while moving images of their swallowing are recorded and evaluated by the speech and language therapy team[40]. This is typically conducted at the child's tertiary hospital (i.e., Great Ormond Street Hospital/Evelina London Children's Hospital).

I will access videofluoroscopic swallow study recordings from hospital databases and analyse them using:

- *Rosenbek PAS* – described in work package 2
- *BaByVFSS Impairment Profile (BaByVFSSImP)* or *Modified Barium Study Impairment Profile (MBSImP)*[36,41,42] - standardised analysis and reporting tools for videofluoroscopic swallow studies. The age of the patient/type of feeding assessed determines which tool to use.

Feeding and swallowing measurement tools used:

A summary of the tools that will be used to characterise PFDs are listed in Table 1. They provide information across the four domains of PFD.

Table 1.

| **Measurement Tool** | **Medical domain** | **Nutritional domain** | **Feeding Skill domain** | **Psychosocial domain** |
|---|---|---|---|---|
| PFD CRF | x | x | x | x |
| CEDAS | | x | x | |
| Pedi-EAT | | | x | x |
| FS-IS | | | | x |
| Videofluroscopic swallow study (Rosenbek PAS, BaByVFSSImp and MBSImP) | x | | x | |

Data analysis:

1. I will analyse feeding and swallowing assessment data, including frame-by-frame secondary analysis of videofluoroscopic swallow study recordings. I will score parent-reported questionnaires according to assessment scoring instructions. I will input assessment data into a password-protected Excel spreadsheet then transfer into SPSS. Data analysis will follow the process described in work package 2.

2. Qualitative parent interviews will be transcribed using Zoom/Teams software (and I will transcribe phone calls). I will check the accuracy of transcripts by re-watching/re-listening to interviews. Interviews will be anonymised and pseudonyms will be used. I will conduct reflexive thematic analysis to analyse interviews using Braun and Clarke's approach[43]. This includes familiarisation with the data, generating initial codes, generating themes, reviewing potential themes, defining and naming themes, and producing a report[43]. A proportion of interviews will be re-coded by a second rater for trustworthiness. I will have a consensus discussion with my supervisors regarding final themes. Results will be shared with the advisory group for sense-checking. Participants will have an opportunity to review findings and ensure interpretation is accurate.

**Synthesis of work packages:**

By completing all three work packages, I will address the research gaps in understanding the occurrence, nature and impact of PFD in ventilator-assisted children under 5 years of age. These findings, combined with the

environmental scan, will provide a holistic picture of children's feeding journeys during and following hospital discharge. This will provide an evidence-based foundation for identifying future research priorities, including i) the development of national guidelines regarding service delivery and practice, and ii) design of intervention tools to improve feeding outcomes of children with long-term ventilation needs.

**6. Dissemination, Outputs, and Anticipated Impacts**

Together with my supervisory team and expert advisory group, I will co-develop accessible outputs for dissemination. I will write a lay summary to stakeholders (e.g. charities, service providers, respiratory and speech and language therapy teams at hospital sites) and study participants. I will share findings through research groups, professional associations and ventilation centres. I will publish in open-access peer-reviewed journals and disseminate outcomes at conferences (see conferences in 'training and development plan'). These activities will be completed with my PPI group. I will create a Twitter/X account to raise awareness of my study and post updates about recruitment, dissemination and outputs. With my advisory group/supervisory team, I will write a best-practice statement regarding service delivery and clinical practice, which will be further developed in my post-doctoral research work.

Potential barriers for future research, development, adoption and implementation:

*Work package 1:*

- Some grey literature/documents may be subject to intellectual property protection - I will reference appropriately when disseminating findings.
- This work package involves developed countries only - findings will only be applicable to nations with similar health economies.
- Cultural considerations in each country may affect their adoption/implementation of research outcomes in similar ways – I will acknowledge this in planning, data analysis and dissemination stages.

*Work package 2:*

- Multicentre studies can be challenging, particularly at sites that are not my main work site - I will obtain a research passport to conduct research across sites. I will adhere to good clinical practice guidelines at all times.
- Other potential barriers include missing/incomplete patient records, recall bias, observer bias and technology issues - these will be mitigated by adhering to a rigorous and prospective missing data handling protocol.

*Work package 3:*

- Potential concerns about data security - the study will be registered with King's College London (KCL) data protection team. The collection, handling and storage of study data will follow KCL data protection guidelines.
- Generalisation of results to the wider population of ventilator-assisted children will be difficult due to the small sample size – this will be acknowledged in data analysis and dissemination stages.

Anticipated impacts:

Research findings will inform current practice of long-term ventilation teams across the UK and has potential to influence future service delivery and healthcare. Outcomes from the environmental scan will provide an overview of local and international centres which will promote discussion about evidence-based care. Findings from work packages two and three will meet current research gaps and provide necessary knowledge to clinicians and academics working in the fields of respiratory medicine and paediatric feeding. These impacts are anticipated to reach service users, the public and services during the fellowship and continue benefitting health and social care sectors after the fellowship.

**7. Project Management**

With guidance from my supervisory team, I will be responsible for managing the timeline and commitments in the fellowship. This includes completing management and governance activities, work packages, training/development and professional development. I will complete the KCL course 'Completing your PhD on time, on budget, and under control' to learn effective time management and project management skills.

I will meet with my primary supervisor every 2 weeks and with the full supervision team monthly. I will attend Guy's and St Thomas' NHS Foundation Trust and King's Health Partners doctoral peer support groups. I will continue to utilise the NIHR Research Support Service for support with research planning, implementation and dissemination.

The primary tasks I will be responsible for (with guidance from my supervisors):

- Ethics submission and data protection registration
- Organising expert advisory group meetings
- Dissemination of study recruitment information
- Recruitment of participants
- Conducting NIHR Equality Impact Assessments to ensure equality, diversity and inclusion for study participants[44]
- Purchasing project equipment
- Planning/development/completion of work packages
- Study data collection and analysis
- Feedback to stakeholders/participants
- Completing my training plan
- Collaboration with PPI groups
- Networking with fellowship collaborators
- Organising attendance at conferences/workshops/presentations
- Managing fellowship research budget (see 'detailed budget' section)
- Disseminating research findings
- Completing PhD thesis

**8. Ethics**

I will abide by my professional body's and KCL's code of conduct. I will obtain ethical approval from the Health Research Authority, KCL and Guy's and St Thomas' NHS Foundation Trust. I will complete data registration with KCL's data protection office. Study procedures will follow ethical guidelines concerning confidentiality, consent, data maintenance and data storage.

Potential ethical issues:

- Clinical assessments will be conducted in families' homes in work package 3. I will follow the university's lone working policy. I will screen out high-risk visits as part of the exclusion criteria. Prior to visiting homes, I will call families to conduct a visit risk assessment (e.g. identify who will be present, animals on site, parking/access from public transport, environmental dangers)). During visits, I will share my location with my supervisors and check in with my supervisors before/after visits using 'My SOS' mobile phone software.
- In the event of a child experiencing clinical deterioration during a home visit, I will use an established hospital trust Standard Operating Procedure to escalate this to appropriate clinical services.
- There are no anticipated risks to families during home visits outside of typical speech and language therapy practice. I have a current Disclosure and Barring Service check and have completed mandatory health and safeguarding training.
- Qualitative interviews with parents/caregivers may result in discussion of sensitive/distressing issues. If participants become upset, I will offer them a break and opportunity to reschedule the interview. Information sheets/consent forms will contain full disclosure about participants' right to withdraw at any time. I will remind distressed participants they can withdraw from the study and signpost them to appropriate self-referral mental health support as needed.
- Recordings from home visits and interviews will be transferred onto an encrypted hard drive and deleted from the camcorder memory card/Zoom/Teams software within 24 hours of being

recorded. Analysis will be completed in a confidential space and will not be shared during dissemination. Information sheets/consent forms will outline processes about confidentiality and recording.

- I am a highly specialist speech and language therapist and will uphold boundaries between my clinical and research roles. During work package 3, I may identify a clinical need for review from the child's community speech and language therapist (e.g. reported recent change in feeding skills). As all participants will have community feeding support, I will signpost the family to discuss concerns with their community team.
- I will ensure there is no conflict of interest when planning, completing and disseminating research studies in the fellowship.

**9. Success Criteria**

Key risks:

- Delays in ethical approval - I will ensure ethical approval is a high priority and allow at least 6 months for approval.
- Delays in completing work packages - the research timetable accounts for leeway in case of unforeseen circumstances.
- Low response rate from stakeholders in work package 1 - my supervisory team have pre-existing research relationships with centres in included countries. I will maximise efforts to use my fellowship collaboration groups/research networks.
- Poor uptake of recruitment in work package 3 - recruitment efforts will be optimised using the NIHR Be Part of Research registry.
- Rescheduling assessments in work package 3 - families may need to reschedule sessions due to illness/hospitalisations/medical appointments. I will allow for flexibility/leeway in the project timeline. If data is collected outside the 3- and 6-month period, reasons will be stated when reporting findings.
- Attrition of participants in work package 3 - I will conduct interviews at two timepoints to facilitate ongoing engagement. Conducting initial interviews at 3 months post-discharge also accounts for potential attrition/inability to collect this data at 6 months post-discharge.

Success criteria:

- Work package 1: Engagement with at least 10 centres; participation from at least 20 health professionals in each country in online survey.
- Work package 2: Accessibility to data at each site.
- Work package 3: Recruitment of at least 10 families.

## 7. Patient & Public Involvement

**Please describe how patients/service users, carers and the public have been involved in developing this proposal**

I used funding awarded from the NIHR Research Design Service Public Involvement Fund to seek feedback about my research proposal from 3 parents of ventilator-assisted children and 3 health professionals. They also provided feedback on my plain English summary.

Research aims: All agreed that the research proposal is valuable and will make a positive impact on children and families. They stated that investigating children's needs after hospital discharge is important and often forgotten in research.

Methods:

- Work package 1: Parents stated the importance of engaging with a range of health professionals who represent both acute and community services.
- Work package 2: The group reported that all three hospitals were suitable sites to collect data due to their large catchment areas and the inclusion of children from a diverse range of demographic, social and cultural backgrounds.
- Work package 3: Parents and health professionals stated that assessing children's PFD characteristics was "needed". They agreed with the clinical assessments chosen. They felt that the video assessment should be conducted in the child's home to be most representative of their eating and drinking skills. They agreed that reporting on videofluoroscopic swallow study results was important. Parents felt conducting a home visit will allow parents/caregivers to build rapport with me and increase the likelihood of them engaging in an open manner in the interviews. Both parents and health professionals felt that exploring the psychosocial impacts on families was "very important" and a valuable part of my research proposal.
- This group felt that the three work packages complemented each other and will provide holistic information about children's feeding journeys and the impacts on their family's lives.

Important considerations:

- Work package 1: Health professionals stated that research should be considerate of the cultural history in different countries, such as respecting first nations groups and acknowledging social health disparities.
- Work package 2: Health professionals discussed potential challenges regarding missing/incomplete patient data.
- Work package 3: Parents emphasised the need to consider the daily stresses of families' lives. Health professionals agreed that methods should involve a large degree of flexibility and convenience for families.

**Please describe the ways in which patients/service users, carers and the public will be actively involved in the proposed research, including any training and support provided**

Following the completion of my PCAF, I have maintained a strong research relationship with parents from the qualitative research study. Three parents have expressed interest in being actively involved in ongoing research activities. These parents will be invited to join an expert advisory group. I will advertise PPI opportunities on the NIHR People in Research website to widen access and participation for greater diversity and inclusion. If access to technology is a barrier for members, face-to-face meetings will take place. Support for individuals with hearing, mobility and visual impairments will be provided. Language translation/interpretation services will be provided.

The expert advisory group will also consist of health professionals working with ventilator-assisted children. I will meet with the advisory group every three months. They will guide wording of study materials (e.g., ethics application, participant consent and information forms) and assist with dissemination of recruitment details and research findings. Parents will be compensated for their time and contribution as per NIHR payment guidelines[45].

I have attended PPI training and will continue to develop my skills and knowledge in this area. I will seek an experienced PPI mentor from my PPI group to guide planning of PPI activities and seek advice about PPI engagement/withdrawal. I will choose a mentor from my PPI group as these parents are regularly involved in research for WellChild charity and have experience in contributing to research/PPI activities about long-term ventilation.

PPI members will be invited to contribute to dissemination of research findings throughout the fellowship. This will include involvement in patient engagement activities, joint presentation of findings at conferences/workshops, and co-writing of articles in relevant publications (e.g. open-access journal articles, newsletters, magazine articles). I will provide information and compensation for relevant training courses that the PPI group may like to attend, such as training for public speaking and writing courses.

I will engage with relevant patient groups and the public during patient engagement events to disseminate project information. I will also raise awareness about my research studies through social media, patient forums and charity networks.

| **If it is considered not appropriate and meaningful to actively involve patients/service users, carers and the public in your proposed research, please justify why** |
| --- |
| Not applicable. |

## 8. Training & Development and Research Support

**Proposed training and development programme**

I will enrol in a PhD at KCL under the Applied Technologies for Clinical Care research division in the Florence Nightingale Faculty of Nursing, Midwifery and Palliative Care (NMPC). I have used the Vitae Researcher Developmental Framework to prepare and structure my plan. Please see my Gantt chart for anticipated time commitments of each course/workshop.

Formal study: I will access courses from the KCL Centre for Doctoral Studies and Health Sciences Doctoral Training Centre which are ideal environments to develop my personal, professional and career development goals. These centres have confirmed that most courses run for 1-2 hours.

Year 1: 1) Research integrity, 2) Collaboration essentials to solve complex problems, 3) Advanced quantitative research methods, 4) Completing your PhD on time, on budget, and under control

Year 2: 1) Managing research data, 2) Advanced qualitative research methods, 3) Qualitative interviewing

Year 3: 1) Writing up the thesis in the sciences, 2) Presenting your research with confidence, 3) Health professional researchers: managing dual careers webinars

Additional training in researcher and academic leader skills:

Year 1:

- Questionnaire design (Social Research Association (SRA)): This will be valuable for planning and implementing the online survey in work package 1.

- Understanding statistical concepts and essential tests (SRA): This will provide foundational knowledge for designing, gathering and interpreting data in all work packages.

- Introduction to Mixed Methods Research (SRA): This will assist me with completing work package 3 which involves both qualitative and quantitative research methods.

- Conducting a literature review (Sage Campus): This will provide me with skills to complete the literature review in work package 1.

- Using PPI to help communicate your research (Cambridge Biomedical Research Centre) – This will develop my skills in collaborating with PPI groups to disseminate research findings throughout and after my fellowship.

- Introduction to qualitative interviewing (SRA): This will prepare me for conducting interviews with parents/caregivers in work package 3.

Year 2:

- The secrets of success in writing and publishing research articles (UK Enhancing the QUAlity and Transparency Of health Research (EQUATOR) Centre Publication School): This will provide me with training to improve my writing/publication skills.

Year 3:

- The essential guide to grant applications (NIHR) – This will develop my grant writing skills in preparation for post-doctoral research funding applications.

- Presentation and pitching skills for researchers (UCL Bloomsbury Postgraduate Skills Network) – This will provide me with skills and knowledge for disseminating research findings during and following the fellowship.

I will continue to attend NIHR Academy training and conferences which provide networking opportunities with other researchers and help identify further training opportunities.

Additional training in practitioner/practice leader skills:

Year 1: Principles for Managing Long-Term Conditions (Faculty of NMPC): A three-month course which will

provide holistic insight into the complexities that need to be considered during research planning, implementation and adoption into health services for patients with long-term conditions.

Year 2: Clinical Leadership (Faculty of NMPC): A three-month course which will provide skills and knowledge for developing my leadership skills as a health professional in clinical academia.

Overseas visit:

In my 3rd year, I will complete a research visit to The Hospital for Sick Children in Toronto. This hospital is Canada's most research-intensive paediatric hospital and the largest paediatric tertiary centre in the country. Their multidisciplinary complex respiratory care clinic supports children with multiple long-term conditions who have ventilation support at home. Their International Learner Program involves observation experiences. I will learn about their model of patient care for ventilator-assisted children. I will share my research outcomes and develop networks with clinical and research teams. I will also explore their research endeavours and foster opportunities for future collaboration.

Professional practice:

I will dedicate 20% of my fellowship to professional practice to maintain clinical competencies and develop skills to become an expert clinical leader in my research area.

I will continue to work at Evelina London Children's Hospital either once a fortnight/month depending on other fellowship commitments at the time. I will maintain required competencies in acute paediatric dysphagia and continue to develop my specialist clinical skills in respiratory medicine (especially long-term ventilation care). I will progress my leadership skills by taking on supervision and mentoring roles within my team.

In order to advance my specialist clinical skills in paediatric ventilation, I will complete a bespoke virtual education and mentorship program from The Hospital for Sick Children in Toronto as part of their International Learner Program. A faculty member will regularly meet with me throughout my fellowship to explore issues in patient care and practice, and create a bespoke online learning program containing relevant modules/courses. Remote mentoring will be provided by a specialist respiratory consultant.

In my PCAF, I met with respiratory consultants across the South Thames region of London and observed their multidisciplinary long-term ventilation clinics. In the DCAF, I will complete similar activities across hospitals in the North Thames region of London on a bimonthly basis. This has been agreed with the Pan-Thames Long-Term Ventilation Programme. This will advance my specialist clinical skills and raise my profile as a clinical academic across these health networks.

I will dedicate the remainder of professional practice time to attending associated professional development. This will include training for BaByVFSSImP, a new standardised assessment tool for interpreting and reporting videofluoroscopic swallow study results. This will provide me with required training for this tool in work package 3. I will also attend a Fiberoptic endoscopic evaluation of swallowing (FEES) training course. FEES is a useful instrumental assessment for evaluating swallowing in ventilator-assisted children. Completing this training will allow me to improve my paediatric feeding skills and specialist knowledge in this area.

Conferences:

I will maintain up-to-date knowledge of emerging research, disseminate research findings and build networks during these conferences. These events will also raise my profile as a clinical academic in my field.

- UK Swallow Research Group (2026, 2028)
- Home Mechanical Ventilation UK Meeting (2026)
- European Society of Swallowing Disorders (2027)
- Feeding Matters conference (2027)
- Royal College of Speech and Language Therapy Conference (2027)

***Primary Doctoral Supervisor***

**PRIMARY DOCTORAL SUPERVISOR 1**

| **Name** | Professor Louise Rose |
|---|---|

| **Institution** | Kings College London |
|---|---|

| **Position** | Research Division Head at the Florence Nightingale Faculty of Nursing, Midwifery and Palliative Care |
|---|---|

| **Description of support provided** |
|---|
| My primary supervisor is Professor Rose who is a Professor of Critical Care Nursing and Research Division Head at the Florence Nightingale Faculty of Nursing, Midwifery and Palliative Care at King's College London. She also holds an honorary Professorship in the Department of Critical Care and at the Lane Fox Respiratory Unit, St. Thomas's Hospital. Professor Rose has authored over 240 peer-reviewed publications, seven book chapters and is a Cochrane Collaboration systematic review author. Her H-index is 55, i10 index is 139, with 9062 citations (April 2024). She has been awarded 114 peer reviewed competitive research grants totalling £20,275,083 with £7,494,884 as chief/primary investigator.<br><br>Professor Rose has research interests that include the trajectory of mechanically ventilated patients from home to intensive care. She has substantial experience in mixed-methods, randomised controlled trial study designs, process evaluations and qualitative interviews.<br><br>Professor Rose is currently supervising seven PhD students, three as primary supervisor (all funded NIHR) and four as supervisory committee member at various universities in the UK and internationally. She is also supervising one NIHR Pre-doctoral Clinical Academic Fellow (PCAF), two Masters students, and one NIHR Advanced Clinical Academic Fellowship. She has supervised to completion a further six PhD students, three post-doctoral fellows, five PCAFs and two UK Masters Dissertation students.<br><br>Professor Rose has 90% dedicated time to research and academic supervision and has ample availability to provide supervision of an exemplary standard. She has provided me with valuable support so far and she is an ideal primary supervisor for my fellowship. |

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***Additional Academic Supervisor(s)***

**ADDITIONAL ACADEMIC SUPERVISOR 1**

| **Name** | Professor Jo Wray |
|---|---|

| **Institution** | University College London |
|---|---|

| **Position** | Senior Research Fellow and Health Psychologist |
|---|---|

| **Description of support provided** |
|---|
| Dr Jo Wray is an experienced health psychologist and leads a programme of mixed-methods psychosocial research in the Heart and Lung directorate at Great Ormond Street Children's Hospital (GOSH). She is a Professor of Child Health Psychology in the Institute of Cardiovascular Science at UCL (Cardiology, Critical Care and Transplantation) and a key element of her GOSH and institutional role is to support and enable clinical academics from a range of nursing and allied health professional backgrounds, as well as develop, mentor and supervise early career researchers. She currently supervises six PhD students from a range of clinical backgrounds, four of whom are NIHR doctoral fellows (she is primary supervisor for two, both are exploring the impact of feeding disorders associated with chronic conditions). Previously she supervised eight PhD students, |

four doctoral clinical psychology students and eight Masters students to completion.

Although Dr Wray works in a different institution to the applicant, provision of regular supervision will be accommodated as she has protected time to provide academic supervision. The proposed project fits well with Dr Wray's current research programme due to her focus on, and interest in, mixed-methods research, particularly with children with complex conditions and their families. She has considerable experience of working with children and families using a wide range of methodologies and is currently a co-applicant on several NIHR-funded projects which include qualitative elements exploring patient and family experiences of illness and impact of treatment, quantitative elements related to clinical and psychosocial data and evidence synthesis.

**ADDITIONAL ACADEMIC SUPERVISOR 2**

| **Name** | Dr Jeanne Marshall |
|---|---|

| **Institution** | Children's Health Queensland and The University of Queensland |
|---|---|

| **Position** | Conjoint Senior Research Fellow |
|---|---|

| **Description of support provided** |
|---|
| Dr Jeanne Marshall is a senior conjoint research fellow at Queensland Children's Hospital and the University of Queensland. Dr Marshall has an expanding research portfolio in the area of paediatric feeding disorder, as well as significant clinical experience as a speech pathologist at Queensland Children's Hospital. Dr Marshall has a career total of 36 publications, 252 citations, a h-index of 10, and a current field-weighted citation index of 2.04 (Scopus, May 2024). Dr Marshall has gained funding for her work via individually-led grant applications, and through mentorship of clinicians and students (career total $395,224). Additionally, Dr Marshall is a consortium representative for Children's Health Queensland on a $6.9 million national project. Dr Marshall has provided/currently provides supervision to 9 research higher degree students. Four of these students have graduated (1x PhD, 2x Masters, 1x clinical doctorate), and 5 students are current PhD students. Supervision has been provided both in Australia, and internationally in the UK and the United States. Dr Marshall has a dedicated 80% of time to research activity, and has availability to support supervision of this project.<br><br>Dr Marshall will provide expertise on paediatric feeding and dysphagia in clinical research. She was my clinical academic mentor and a valuable part of my supervision team during my PCAF. Her contribution to my PCAF has been integral to the publishing and dissemination of our research outcomes. She has a passion for research translation and building research capacity, and is an inspiration for clinical academic speech and language therapists. |

***Clinical Supervisor(s)***

**PRACTICE SUPERVISOR 1**

| **Name** | Alexandra Stewart |
|---|---|

| **Institution** | Great Ormond Street Hospital |
|---|---|

| **Position** | Speech and Language Therapist and current DCAF Fellow |
|---|---|

| **Description of support provided** |
|---|
| Alex has been working as a speech and language therapist in the field of paediatric feeding difficulties for over 20 |

years. She is based at Great Ormond Street Hospital and is due to complete an NIHR-funded Clinical Doctoral Research Fellowship in July 2024. She has provided clinical mentorship to three speech and language therapists undertaking PCAFs (2 completed, 1 current) and supervised six MSc research projects to completion. Alex is chair of the Royal College of Speech and Language Therapists paediatric dysphagia clinical excellence network, scientific chair of the UK Swallowing Research Group, has been involved in a number of professional body working groups and founded the Paediatric Dysphagia Speech and Language Therapists in Research network. She has experience of leading and innovating within clinical practice. She is currently establishing use of high-resolution impedance manometry in pharyngeal swallow assessment in the UK. Alex will provide supervision once every two months which will include development of clinical skills, such as use of fibreoptic endoscopic evaluation of swallowing, support with clinical aspects of the research project and mentorship to implement changes in clinical practice. In addition, she is extremely well-placed to strengthen access to a network of researchers and clinicians in the field, provide guidance on managing competing aspects of the fellowship and champion my development as a clinical academic.

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**Collaborations**

*WellChild Charity:* WellChild is a national charity for children with complex health needs who are cared for at home. I collaborated with WellChild in my PCAF and have a strong clinical academic relationship with their team. I will have access to their family engagement team, family network and multidisciplinary team who are specialised in long-term ventilation. WellChild charity has agreed to share research study information on social media and disseminate research findings. I will plan ongoing opportunities for networking and research endeavours for collaborations in future post-doctoral work.

*Pan-Thames Paediatric Long-Term Ventilation Programme:* This programme is commissioned by NHS England London and geographically covers both the South and North Thames Paediatric Operational Delivery Networks (ODNs). I collaborated with the South Thames Long-Term Ventilation Paediatric Network in my PCAF and have a strong clinical academic relationship with their team. The Pan-Thames programme has agreed to share research study information through their network, recruit participants and disseminate research findings. I will visit respiratory clinics that are part of the North Thames Paediatric ODNs to advance my specialist knowledge. Working in collaboration with this team provides me with a robust group of health professionals and managers to collaborate with in current and future research endeavours.

*Acute paediatric speech and language therapy teams at Evelina London Children's Hospital and Great Ormond Street Hospital:* These two teams provide specialist tertiary care to ventilator-assisted children with feeding and swallowing needs. I will collaborate with them during completion of work package three to screen for eligible participants. I will also disseminate my research findings to these teams and their networks.

*Respiratory teams at Evelina London Children's Hospital and Great Ormond Street Hospital:* These two large specialist tertiary teams contain respiratory consultants, specialist nurses and allied health professionals. Collaboration with these teams allows me to share research study information, recruit participants and disseminate findings through their networks. I will raise my profile as a clinical academic across these hospitals and their wider health networks in England.

*Multidisciplinary Complex Respiratory Care Clinic at The Hospital For Sick Children in Toronto:* This hospital's mission statement involves partnering globally to improve children's health outcomes in care, research and education. I will complete an overseas research visit as part of the hospital's International Learner Program which involves observational experiences and a virtual education/mentoring program. This will advance my clinical and research skills in this area. I anticipate this collaboration leading to ongoing networking, raising of my profile as a clinical academic, and future opportunities for international research.

*Expert Advisory Group:* This group will involve the following health professionals whom have agreed to collaborate with me during this fellowship –

- *Alison Felton (Highly specialist speech and language therapist):* Alison works at Alder Hey Hospital and specialises in the assessment and management of neuromuscular conditions and long-term ventilation. She completed an NIHR PCAF in 2023 and focused on the topic of spinal muscular atrophy during this fellowship. Her knowledge and skills in the field of ventilation care in both clinical and research fields will be integral to the planning and implementation of my research proposal.
- *Respiratory consultants (Dr Jane Heraghty and Dr Simona Turcu):* Dr Heraghty and Dr Turcu are paediatric respiratory consultants at Evelina London Children's Hospital. Dr Heraghty established the hospital's respiratory and long-term ventilation service. She collaborated with me in my PCAF and we have a strong clinical academic relationship. Dr Heraghty and Dr Turcu will contribute valuable expertise to the expert advisory group.

**Host Organisation Support Statement**

The following statement of support should be composed jointly by the Heads of Department / Senior Managers of the employing host organisation and the partner organisation.
The Heads of Department / Senior Managers of these organisations will be able to add the statement of support to the form once they have confirmed participation. Invitations to participate are sent by the applicant via the 'Participants and Signatories' section of the form.

**Statement of support**

On behalf of the acute speech and language therapy (SLT) service at Guy's and St Thomas' NHS Foundation Trust (GSTT) and King's College London (KCL), we are delighted to support Sabrena's application for an NIHR Doctoral Clinical and Practitioner Academic (DCAF) fellowship. We confirm that, if the award is made, Sabrena will be integrated into both departments enabling her to conduct clinical and research development activity appropriate to the Fellowship.

There is a strong history of collaboration and partnership between the Evelina, Neurosciences department of the Guy's & St Thomas' NHS Foundation Trust (GSTT) and King's College London (KCL) Faculty of Life Sciences and Medicine (FoLSM), both committed to supporting Sabrena throughout her clinical academic development. Evelina has previously successfully supported 8 PCAF and 5 DCAF amongst allied health professionals.

The School of Life Course & Population Sciences (in FoLSM) unites experts in women's and children's and population health. Units of assessment within the faculty were among top performers in the 2021 Research Excellence Framework. For Allied Health Professions, Dentists and Pharmacy KCL was ranked first out of 89 submissions in proportion overall percentage ranked at 4*, GPA and Power rankings. KCL is 7th in the UK's top universities and 35th in the world's top universities in the QS World University Rankings (2023).

The acute SLT service has a growing reputation for supporting staff in research development whilst developing their clinical practice. At both a Trust and service level the staff in the service regularly collaborate with clinical academics to conduct lectures, courses, and Continued Professional Development activities to ensure our therapists are well informed and implementing the most up-to-date evidenced-based practice. We work with HEI's to support clinical placements and Sabrena is a practice educator. Sabrena has also participated in a wide range of service development and audit projects including the design and production of an online dysphagia training module for nurses, a vocal fold palsy audit and the development of an inpatient SLT service for patients with LTV. Sabrena was also involved in evaluating a survey of parents/caregivers care experience of the inpatient SLT service. This project is approved by the service and her proposed area of study fits in with Sabrena's current clinical commitments and interest. It will be of benefit to the SLT Service for Sabrena to undertake this research project.

The KCL Florence Nightingale Faculty of Nursing, Midwifery and Palliative Care ('The Faculty') is an integral partner within KHP and contributes to the interdisciplinary research centres in delivering high quality research. For the last three years, King's has been number 2 in the world and number 1 in Europe for Nursing (QS world

rankings). The 2021 Research Excellence Framework (REF2021) placed King's applied and allied health research 1st for all of the following: overall proportion of research rated 4*, research papers rated 4*, impact rated 4*, environment rated 4*(joint), Grade Point Average and Power. REF2021 rated 100% of our applied and allied health research environment, 86% of our impact and 70% of our research overall as world-leading. Over 95% of our research overall was rated as world-leading or internationally excellent.

The Faculty's applied clinical and health research develops, understands, implements, and evaluates interventions, therapies, and care across our fields to build knowledge, lead policy and practice and improve outcomes for patients and those important to them. We work with local services and our partner NHS Trusts, lead national studies supported by NIHR, UKRI, EU and NIH and globally, and major research programmes in Africa and Asia. The Faculty where Sabrena will be based, has a dedicated research support team led by Professor Glenn Robert, (Vice Dean for Research and Impact). This team leads on implementation of the Faculty's research strategy and provides the infrastructure required to deliver tailored training to individual researchers. Sabrena will also have access to all relevant modules and training provided via KCL and the Specialist Biomedical Research Centres based within KHP, in addition to monthly grant writing workshops and research division meetings offered by the Faculty.

Sabrena will work daily alongside Faculty researchers who are tackling some of the most enduring health issues. If successful, Sabrena will become a member of the Applied Technologies in Clinical Care division. Faculty academics have research networks through their commitments with national and international committees. Working alongside these senior investigators within the Faculty and wider KHP will give Sabrena the opportunity to learn more about designing and conducting large and complex studies and provides support networks and mentorship for ensuring her findings have impact and are translated into practice and policy. Sabrena will learn more about translating and disseminating research findings to directly inform practice and policy. All doctoral students in the Faculty are actively involved with developing their research's impact, through engagement with clinicians, managers, service-users and policy-makers at all stages, to prepare for the translation and implementation of their research, from its original inception through to dissemination.

GSTT demonstrates a commitment to progressive academic study and clinical academic roles for AHPs. The trust have appointed the first AHP Research Lead for the organisation, who has a position on the Trust R&D board. The Trust has also just published its new AHP strategy with research as a stated priority, including supporting clinical academic careers, hosting of clinical academic posts and commitment to supporting clinical academic training.

As an organisation GSTT is fully committed to improving the experiences of all staff and patients, by providing inclusive healthcare to patients, pledging to be an inclusive workplace and promising workplace equality. The GSTT R&D strategic choices paper has prioritised equality, diversity and inclusive approaches to offer all staff the opportunity for involvement in research design and delivery, and to meet the highest standards of research integrity. KCL has both a Race Equality Charter and Athena Swan award and, through its Race Equality and Athena Swan Action Plans, is committed to building an anti-racist University and to transforming and supporting gender and race equality in research.

## 9. Detailed Budget

**The finance section should provide a breakdown of costs associated with undertaking the research as described in the proposal.**

**1**

| **Staff type** | Lead Applicant |
|---|---|

| **Name** | Sabrena Lee |
|---|---|

| **Employing organisation** | Guy's and St Thomas' NHS Foundation Trust |
|---|---|

| **Grade** | 7 |
|---|---|

| **Starting point on scale** | 08 |
|---|---|

| **Increment Date** | 28/08/2024 |
|---|---|

| **Justification of Costs** |
|---|
| These costs have all been provided by the Guy's and St Thomas' Hospital NHS Foundation Trust Research and Development Finance team.<br><br>£289 in 'other costs' is the annual apprentice levy cost. |

**Details of Posts and Salaries Costs**

| | **Total (£)** |
|---|---|
| Salary (£) | 50056 |
| Geographical Weighting (£) | 7746 |
| Other allowances (£) | 289 |
| Superannuation (£) | 8312 |
| Nat. Insurance (£) | 7330 |
| **Current Annual Costs (£)** | **73733** |

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**Direct Costs**
These are the costs that are specific to the research, which will be charged as the amount actually spent and can be supported by an audit record.

**Staff Type:** (Lead Applicant)-Miss Sabrena Lee
**Type of cost:** NHS
**% Full-time on this research:** 100.0
**Total months on this research:** 36.0

**Justification of Cost:** Geographical weighting / Other allowances include the following: London weighting= £7746, apprentice levy = £289. These costs have all been provided by the Guy's and St Thomas' Hospital NHS Foundation Trust Research and Development Finance team.

| | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| Basic Salary | 50056 | 50056 | 50056 | 150168 |
| National Insurance | 7330 | 7330 | 7330 | 21990 |
| Superannuation | 8312 | 8312 | 8312 | 24936 |
| Geographical weighting / Other allowances (if applicable) | 8035 | 8035 | 8035 | 24105 |

| Totals | Applicant | Support Staff | Shared Staff | Total Cost (£) |
|---|---|---|---|---|
| Total HEI | 0 | 0 | 0 | 0 |
| Total NHS | 221199 | 0 | 0 | 221199 |
| Total Commercial | 0 | 0 | 0 | 0 |
| Total Other | 0 | 0 | 0 | 0 |

**Travel, Subsistence and Conference Fees**

**Type of cost:** NHS
**Description:** Conference Fees
**Conference:** True
**Item Description:** Conference Fees
**Justification of Cost:** I will attend a range of conferences that will inform me about up-to-date emerging research and provide opportunities for research networking and dissemination. This includes conferences focused on paediatric feeding, including 1. UK Swallow Research Group (£500), 2. European Society of Swallowing Disorders (£250), 3. Feeding Matters (£200), and 4. Royal College of Speech and Language Therapy conference (£100). I will also attend quarterly Paediatric Dysphagia Clinical Excellence Meetings (£120). I will also attend conferences focused on respiratory medicine and long-term ventilation including the Home Mechanical Ventilation UK meeting (£100). Costs for travel and accomodation to these conference is listed under 'journey costs' (and 'conference' is ticked).

| | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| | 390 | 390 | 390 | 1170 |

**Type of cost:** NHS
**Description:** Journey Costs
**Conference:** True
**Item Description:** Travel and accomodation to conferences
**Justification of Cost:** The following conferences will require travel expenses: 1. UK Swallow Research Group (within the UK - £100) and 2. European Society of Swallowing Disorders (within Europe. - £450). I will be attending the UK Swallow Research Group conference in Year 1 and 3. Taxi fees from and to the airport for the European Society of Swallowing Disorders has also been budgeted (£100).

| | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| | 100 | 550 | 100 | 750 |

**Type of cost:** NHS
**Description:** Subsistence
**Conference:** True
**Item Description:** Subsistence for Conference Attendance
**Justification of Cost:** I will require funding for subsistence for the UK Swallow Research Group conference (£300 for accommodation and meals) and the European Society of Swallowing Disorders conference (£400 for accommodation and meals). I will attend the UK Swallow Research Group conference in years 1 and 3. I will attend the European Society of Swallowing Disorders conference in year 2.

| | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| | 300 | 400 | 300 | 1000 |

**Type of cost:** NHS
**Description:** Journey Costs
**Conference:** False
**Item Description:** Work package 2 journey costs for lead applicant
**Justification of Cost:** For work package 2, I will travel to two hospital sites in London (once a week) for 12 weeks. I will travel by underground and each return journey will cost me £8.80. Therefore a total of £105.60 has been requested for my journey costs. As Alder Hey Hospital is in Liverpool, I will also require costs for two trips to Liverpool. This includes return train tickets to travel to Liverpool (2x £150) and accommodation for 2 nights (2x £200).

| | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| | 0 | 806 | 0 | 806 |

**Type of cost:** NHS
**Description:** Journey Costs
**Conference:** False
**Item Description:** Work package 3 journey costs for lead applicant
**Justification of Cost:** In this work package, I will travel to patients/families' homes in order to conduct assessments and interviews. A return train journey from my home to a borough outside of London typically costs approximately £50 return. I anticipate a maximum sample size of 30 patients therefore request £1500 for my anticipated travel costs.

| | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| | 0 | 1500 | 0 | 1500 |

**Type of cost:** NHS
**Description:** Journey Costs
**Conference:** False
**Item Description:** Work package 3 journey costs for lead applicant
**Justification of Cost:** In this work package, I will travel to two hospital sites to meet and recruit patients. I anticipate I will visit each hospital site three times over the course of the study. Each return underground journey will cost £8.80. Therefore I request £53 for these journey costs.

| | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| | 53 | 0 | 0 | 53 |

**Type of cost:** NHS
**Description:** Journey Costs
**Conference:** False
**Item Description:** Travel costs associated with overseas research visit
**Justification of Cost:** In my 3rd year, I will complete a research visit to The Hospital for Sick Children in Toronto. This hospital is Canada’s most research-intensive hospital and the largest paediatric tertiary centre in the country. Their International Learner Program involves observation experiences. The minimum duration for an observation visit as part of the hospital's International Learning Program is 5 days. I will therefore visit the hospital for 5 days. These journey costs include return flights (£1000), accommodation for 6 nights (£1500), taxi fees to and from the airport (£100) and meals (£210 (£35/day)). Current return flights to Canada in March 2025 are approximately £800, therefore the cost requested above includes estimated cost for travel in 2028.

| | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| | 0 | 0 | 2810 | 2810 |

| | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| Travel, Subsistence and Conference Fees Total | 843 | 3646 | 3600 | 8089 |

**Equipment**

**Type of cost:** NHS
**VAT relief to be applied:** False
**Description:** Encrypted hard drive
**Price excluding VAT £:** 200
**Expected life time:** 36
**Justification of Cost:** I will use an encrypted hard drive to store all research study data for work packages 1, 2 and 3. This will ensure that confidential data is stored in an appropriate manner, in line with data protection regulation guidelines. This will also ensure that research data is maintained in a safe and secure way. The encrypted hard drive will be stored securely in a lockable drawer and the key will be accessed by the lead applicant only.

| | Year 1 (exc VAT) | Year 2 (exc VAT) | Year 3 (exc VAT) | Total Cost (£) (exc VAT) |
|---|---|---|---|---|
| | 200 | 0 | 0 | 200 |

**Type of cost:** NHS
**VAT relief to be applied:** False
**Description:** Video camcorder, tripod and SD memory card
**Price excluding VAT £:** 150
**Expected life time:** 36
**Justification of Cost:** In work package 3, I will be conducting clinical assessments on participants during home visits. I will require a video camcorder for this process. I will also use this camcorder to record phone interviews in this work package.

| | Year 1 (exc VAT) | Year 2 (exc VAT) | Year 3 (exc VAT) | Total Cost (£) (exc VAT) |
|---|---|---|---|---|
| | 0 | 150 | 0 | 150 |

| | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| Equipment Total | 200 | 150 | 0 | 350 |

**Consumables**
No costs have been added

**Patient and Public Involvement**

**Description:** PPI costs
**Type of cost:** NHS
**Please describe fully:** PPI costs
**Justification of Cost:** PPI members will be a valuable part of the expert advisory group. I will also have a PPI mentor during this fellowship. NIHR PPI payment guideline advises researchers to compensate each PPI member £50 for a two hour activity. I will likely have 3x PPI members in my expert advisory group. We will meet online every 3 months during the fellowship. Therefore, a requested cost of £600/per year is requested to cover the PPI members' attendance at expert advisory group meetings. Additional costs associated with training and contribution to dissemination of research findings (including conference fees) are included. This will be costed at £600 per year for the PPI group to use. This total cost may not be utilised every year and will also be used for organising services for PPI members with hearing, mobility or visual impairments (such as sign language interpreters, talk-to-type conversion) and for interpreters. Any costs that are not utilised from each year will be rolled over for use in the next year. This will ensure all PPI costs are able to be compensated for as PPI involvement/engagement is a valuable part of this fellowship.

| | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| | 1100 | 1100 | 1100 | 3300 |

| | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| Patient and Public Involvement Total | 1100 | 1100 | 1100 | 3300 |

**Other Direct Costs**

**Description:** Health Research Authority Registration Fees
**Type of cost:** NHS
**Please describe fully:** Health Research Authority Registration Fees
**Justification of Cost:** My host organisation, GSTT, has advised that each work package requires a Health Research Authority registration fee of £100. Therefore, I have calculated the following costs for three work packages.

| | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| | 100 | 100 | 100 | 300 |

**Description:** Archiving
**Type of cost:** NHS
**Please describe fully:** Archiving
**Justification of Cost:** My host organisation, GSTT, has advised including archiving costs at a baseline cost of £500 per site. This will be required for work package 2 and 3. Work package 2 takes place over 3 sites and work package 3 takes place over 2 sites.

| | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| | 0 | 1500 | 1000 | 2500 |

**Description:** Interpreting Services
**Type of cost:** NHS
**Please describe fully:** Interpreting Services
**Justification of Cost:** In work package 3, I may require translation of research participant material into other languages. I may also require an interpreter during interviews with parents/caregivers. The cost of translating one page (500 words) is between £50-£100. The average day rate of an interpreter starts from £100 and their may also include travel time.

| | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| | 0 | 500 | 0 | 500 |

| | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| Other Direct Costs Total | 100 | 2100 | 1100 | 3300 |

**Training And Development**

**Type of cost:** HEI
**Description:** Applicant PhD Tuition Fees
**Item Description:** King's College London PhD Fees
**Justification of Cost:** The estimated tuition fees for a full-time home PhD for each year from 2025-2028 have been provided by the research support manager at King's College London Nightingale Faculty of Nursing, Midwifery and Palliative Care.

| | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| | 8427 | 8933 | 9469 | 26829 |

**Type of cost:** NHS
**Description:** Workshops
**Item Description:** Questionnaire Design
**Justification of Cost:** This course is run by the Social Research Association. It takes place over two half-day sessions and will provide essential training for planning and implementing the online survey in work package 1.

| | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| | 220 | 0 | 0 | 220 |

**Type of cost:** NHS
**Description:** Workshops
**Item Description:** Understanding statistical concepts and essential tests
**Justification of Cost:** This one-day course is run by the Social Research Association. It will provide me with foundational knowledge for designing, gathering and interpreting data in all work packages.

| | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| | 220 | 0 | 0 | 220 |

**Type of cost:** NHS
**Description:** Workshops
**Item Description:** Introduction to mixed methods research
**Justification of Cost:** This one-day course will provide me with valuable skills and knowledge for the planning and implementation of work package 3 which involves both qualitative and quantitative research methods. It is run by the Social Research Association.

| | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| | 220 | 0 | 0 | 220 |

**Type of cost:** NHS
**Description:** Short Courses
**Item Description:** Principles for Managing Long-Term Conditions
**Justification of Cost:** This short course is hosted by the Florence Nightingale Faculty of Nursing, Midwifery and Palliative Care at King's College London. It is a three-month course which will provide me with holistic insight into the complexities that need to be considered during research planning, implementation and adoption into health services for patients with long-term conditions.

| | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| | 3100 | 0 | 0 | 3100 |

**Type of cost:** NHS
**Description:** Workshops
**Item Description:** BaByVFSSImp Training
**Justification of Cost:** The BaByVFSSImp assessment tool is a standardised assessment tool for interpreting and reporting videofluoroscopic swallow study assessments. This tool will be released in September 2024. The cost included below is estimated based on the current cost of a similar training workshop (MBSImP) by the same organisation (Northern Speech Services).

| | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| | 500 | 0 | 0 | 500 |

**Type of cost:** NHS
**Description:** Short Courses
**Item Description:** Clinical Leadership
**Justification of Cost:** This three-month course is run by the Florence Nightingale Faculty of Nursing, Midwifery and Palliative Care at King's College London. It will provide me with the skills and knowledge for developing my leadership skills as a health professional in clinical academia.

| | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| | 0 | 3100 | 0 | 3100 |

**Type of cost:** NHS
**Description:** Workshops
**Item Description:** FEES training course
**Justification of Cost:** Imperial College London hosts a Fiberoptic endoscopic evaluation of swallowing (FEES) training course. FEES is an instrumental assessment used for the evaluation of swallowing and is a useful tool for ventilator-assisted children. Completing this training will allow me to improve my paediatric feeding skills and specialist knowledge in this area.

| | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| | 0 | 0 | 500 | 500 |

**Type of cost:** NHS
**Description:** Overseas Research Visits
**Item Description:** Overseas Research Visit to The Hospital for Sick Children in Toronto
**Justification of Cost:** In my 3rd year, I will complete a research visit to The Hospital for Sick Children (SickKids) in Toronto. This hospital is Canada’s most research-intensive hospital and the largest paediatric tertiary centre in

the country. Their International Learner Program (ILP) involves observation experiences. Their multidisciplinary complex respiratory care clinic provides care to children with MLTC-M who require ventilation support at home. I will learn about their model of patient care for ventilator-assisted children, especially in the management of feeding and swallowing difficulties. I will share my research outcomes and develop my relationship with the clinical and research teams. I will also explore their current research endeavours and foster opportunities for future collaboration. The minimum duration for an observation visit as part of the hospital's International Learning Program is 5 days. I will therefore visit the hospital for 5 days. The budgeted cost below includes the hospital's application fee for the International Learner Program ($200CAD / £116).

| | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| | 0 | 0 | 116 | 116 |

**Type of cost:** NHS
**Description:** Workshops
**Item Description:** Introduction to qualitative interviewing
**Justification of Cost:** This one-day online course provided by the Social Research Association will provide me with necessary skills and knowledge to prepare and conduct qualitative interviews in work package 3.

| | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| | 220 | 0 | 0 | 220 |

**Type of cost:** NHS
**Description:** Workshops
**Item Description:** The secrets of success in writing and publishing research articles
**Justification of Cost:** This four-day course is hosted by the UK EQUATOR Centre Publication School. It is designed for early-career researchers and students. It will provide me with the necessary skills and knowledge to improve my planning, writing and publishing of my research study findings throughout the fellowship.

| | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| | 0 | 150 | 0 | 150 |

| | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| Training And Development Total | 12907 | 12183 | 10085 | 35175 |

**Total Direct Costs**

| | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| Total Direct Costs | 88883 | 92912 | 89618 | 271413 |

**Indirect costs**

**HEI Indirect costs**
No costs have been added

**Commercial Indirect Costs**
No costs have been added

**Other Partner Organisation Indirect costs**
No costs have been added

| | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| Total Indirect Costs | 0 | 0 | 0 | 0 |

**Research Costs Requested From Funder**

| | Direct costs | Indirect costs/Overheads | Amount requested |
|---|---|---|---|
| Total Higher Education Institution Costs (80%) | 0 | 0 | 0 |
| Total Higher Education Institution Costs (100%)s | 26829 | 0 | 26829 |
| Total Commercial Costs | 0 | 0 | 0 |
| Total Other Parther Organisation Costs | 0 | 0 | 0 |
| Total NHS Research Costs | 244584 | 0 | 244584 |
| **Total Research Costs Requested From Funder** | 271413 | 0 | 271413 |

**Summary**

| | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| **Salaries** | **73733** | **73733** | **73733** | **221199** |
| **Travel-subsistence-and-conference-fees** | **843** | **3646** | **3600** | **8089** |
| **Equipment** | **200** | **150** | **0** | **350** |
| **Consumables** | **0** | **0** | **0** | **0** |
| **Patient-and-Public-Involvement** | **1100** | **1100** | **1100** | **3300** |
| **Other-Direct-Costs** | **100** | **2100** | **1100** | **3300** |
| **Training And Development** | **12907** | **12183** | **10085** | **35175** |
| **HEI-Indirect-costs** | **0** | **0** | **0** | **0** |
| **Commercial-Indirect-Costs** | **0** | **0** | **0** | **0** |
| **Other-Partner-Organisation-Indirect-costs** | **0** | **0** | **0** | **0** |
| **Total** | **88883** | **92912** | **89618** | **271413** |

**NHS Support Costs**

| Cost Item: | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| NHS Support Costs | 0 | 0 | 0 | 0.0 |

| **Justification:** |
|---|
| **NHS Support Costs:** No NHS support costs have been added as all study methods will be carried out by the lead applicant and no NHS support staffing is required. |

| | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| NHS Support Costs Total | 0 | 0 | 0 | 0 |

| | |
|---|---|
| **Have you discussed and agreed these costs with the Lead Network?** | Yes |

**NHS Treatment Costs**

| Cost Item: | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| NHS Treatment Costs | 0 | 0 | 0 | 0.0 |

| Justification: |
|---|
| **NHS Treatment Costs:** Work packages include 1) an environmental scan of health professionals and organisations, 2) a retrospective cohort study of patient records, and 3) a prospective case series study including clinical assessments and qualitative interviews. No NHS treatment costs have been added as all research methods will be carried out by the lead applicant and participants will not receive any treatment in these research studies. |

| | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|
| NHS Treatment Costs Total | 0 | 0 | 0 | 0 |

| | |
|---|---|
| **Have you discussed and agreed the NHS costs with the Lead Trust?** | Yes |

| Is the patient/service user care being provided different from the usual standard treatment for the condition? |
|---|
| N/A |

**Usual Treatment Costs**
No costs have been added

**Excess Treatment costs/(savings)**

| | Cost per patient(£) | Year 1 | Year 2 | Year 3 | Total Cost (£) |
|---|---|---|---|---|---|
| Total NHS Excess Treatment costs/(savings) | 0 | 0 | 0 | 0 | 0 |

| | Total (£) |
|---|---|
| NHS Support Costs requested from networks | 0 |
| NHS Excess Treatment costs/savings | 0 |
| TOTAL NHS SUPPORT AND TREATMENT COSTS | 0 |

**Total Funding Required (including non-research costs)**

| | Total (£) |
|---|---|
| Total Research Costs Requested from Funder | 271413 |
| Total NHS (Support and Excess Treatment) Costs | 0 |
| Total Funding Required | 271413 |

## 10. Uploads

Please upload the following documents, if necessary.

When uploading a document to your application, please ensure that it is in an appropriate file format. It is not possible to upload JPEG images, and with the exception of the SoECAT Form, which should be uploaded as an Excel spreadsheet, all files should be uploaded in Adobe PDF or Word DOC/DOCX format. Please refer to the Guidance Notes for further information on the attachments required.

(References - Appendix I References.pdf) is included as an appendix within this file.

(Research Timetable - Gantt Chart SL.pdf) is included as an appendix within this file.

(SoECAT Template - SoECAT Tool A Sabrena Lee_authorised_EF_4.6.2024.xlsx) is included as an appendix within this file.

## 11. Participants and Signatories

**Participants**
You are required to supply the name and email address (if not already registered on the ARAMIS application system) of each required participant. Everyone included in this section will be acting as a 'participant' in your application and must agree to be named as such.

By confirming participation, participants are acknowledging their involvement and input into this application, and are agreeing to be involved, as indicated, in any resultant fellowship.

Once the 'Administrative Authority or Finance Officer' participant has confirmed their participation, they will be able to access the 'Detailed Budget' section of the application form. You should collaborate with your 'Administrative Authority or Finance Officer participant to ensure that this section is completed satisfactorily.

You must ensure that all participants are happy for your application to be submitted before you submit it.

It is permissible for a single individual to serve as both a signatory and a participant, but they will need to be added separately under each heading.

**Named participants must confirm their participation on your application before you will be able to select the 'SUBMIT' option. Please refer to the Submission Process Flow Diagram and the Applicant Guidance Notes for further information.**

***Primary Doctoral Supervisor***

Doctoral Primary Supervisor Participant 1

| | |
|---|---|
| **Title** | Professor |
| **Forename(s)** | Louise |
| **Surname** | Rose |
| **Position** | Professor in Critical Care Nursing |

***Additional Academic Supervisor(s)***

Supervisor Participant 1

| | |
|---|---|
| **Title** | Dr |
| **Forename(s)** | Jeanne |
| **Surname** | Marshall |
| **Position** | Conjoint Clinical Research Fellow |

Supervisor Participant 2

| | |
|---|---|
| **Title** | Professor |
| **Forename(s)** | Jo |
| **Surname** | Wray |
| **Position** | Health Psychologist and Senior Research Fellow |

***Practice Supervisor(s)***

Practice Support Participant 1

| | |
|---|---|
| **Title** | Ms |
| **Forename(s)** | Alexandra |
| **Surname** | Stewart |
| **Position** | Speech and Language Therapist |

***Administrative Authority or Finance Officer***

Administrative Authority or FO Participant

| | |
|---|---|
| **Title** | Mr |
| **Forename(s)** | Ernie |
| **Surname** | Finch |
| **Position** | Senior Financial Analyst |

**Signatories**
You are required to supply the name and email address (if not already registered on the ARAMIS application system) of each required 'Head of Department / Senior Manager' signatory. Once their contact details have been entered below, these signatories can be invited to log in and confirm their participation in your application.

Once the nominated signatories have confirmed their participation, they can access and complete the required 'Host Organisation Statement of Support' in the 'Training & Development and Research Support' section of the application form.

Your signatories must confirm their participation in your application and complete the section they are responsible for before you can select the 'submit' option.

**Signatories must also approve your application after you select the 'submit' option but BEFORE the application submission deadline.**

**Signatory approval from your 'Head of Department / Senior Manager' signatories will result in the application being fully submitted to the NIHR. All parties (applicant, participants and signatories) will be notified of this via an automated system-generated email.**

**The NIHR will not accept your application unless fully approved by your signatories before the 1 p.m. deadline on Tuesday 11th June 2024.**

***Heads of Departments / Senior Managers***

Head of Department or Senior Manager 1

| | |
|---|---|
| **Title** | Professor |
| **Forename(s)** | Ming |
| **Surname** | Lim |
| **Position** | Head of Service |

Head of Department or Senior Manager 2

| | |
|---|---|
| **Title** | Professor |
| **Forename(s)** | Glenn |
| **Surname** | Robert |
| **Position** | Chair in Healthcare Quality & Innovation |

## 12. Acknowledgement, Review and Submit

| **Conflict checks:**<br>Please declare any conflicts or potential conflicts of interest that you may have in undertaking this research, including any relevant, non-personal & commercial interest that could be perceived as a conflict of interest. |
|---|
| I have no conflicts or potential conflicts of interest to declare. |

**By completing this form, the programme reserves the right to share, in confidence, details of your application with other parts of the NIHR and with the Department of Health and Social Care.**

| **Agreement to Terms and Conditions:**<br><br>I confirm that I have secured all necessary licences and ethical approvals in relation to the research and will abide by the terms of those licences and approvals in carrying out the research.<br><br>I confirm that I have read the NIHR Carbon Reduction Guidelines and, where possible, have taken steps to reduce the carbon emissions generated by this research.<br><br>I confirm that all material within this application, with the exception of sections that other participants are required to complete, is my own work and I have taken steps to ensure that no parts of the application have been inappropriately plagiarised.<br><br>I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time lay down according to the NIHR research governance framework.<br><br>I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time laydown according to the UK Policy Framework for Health and Social Care Research.<br><br>In completing this, I declare that, to the best of my knowledge, the information provided in this application, is true, accurate and complete. I have read the Guidance Notes and agree to accept the process by which an application is assessed and agree to abide by the conditions under which an award may be granted. |
|---|
| ☒Confirmed |

# Appendices

1. Appendix I References.pdf
2. Gantt Chart SL.pdf
3. SoECAT Tool A Sabrena Lee_authorised_EF_4.6.2024.xlsx

# Appendix #1: Appendix I References.pdf

**Appendix I: References**


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2. Lee S, Marshall J, Clarke M, Smith CH. Feeding and swallowing outcomes of children receiving long-term ventilation: A scoping review protocol. *PLoS One*. 2024;19(2):e0287872. doi:10.1371/journal.pone.0287872.
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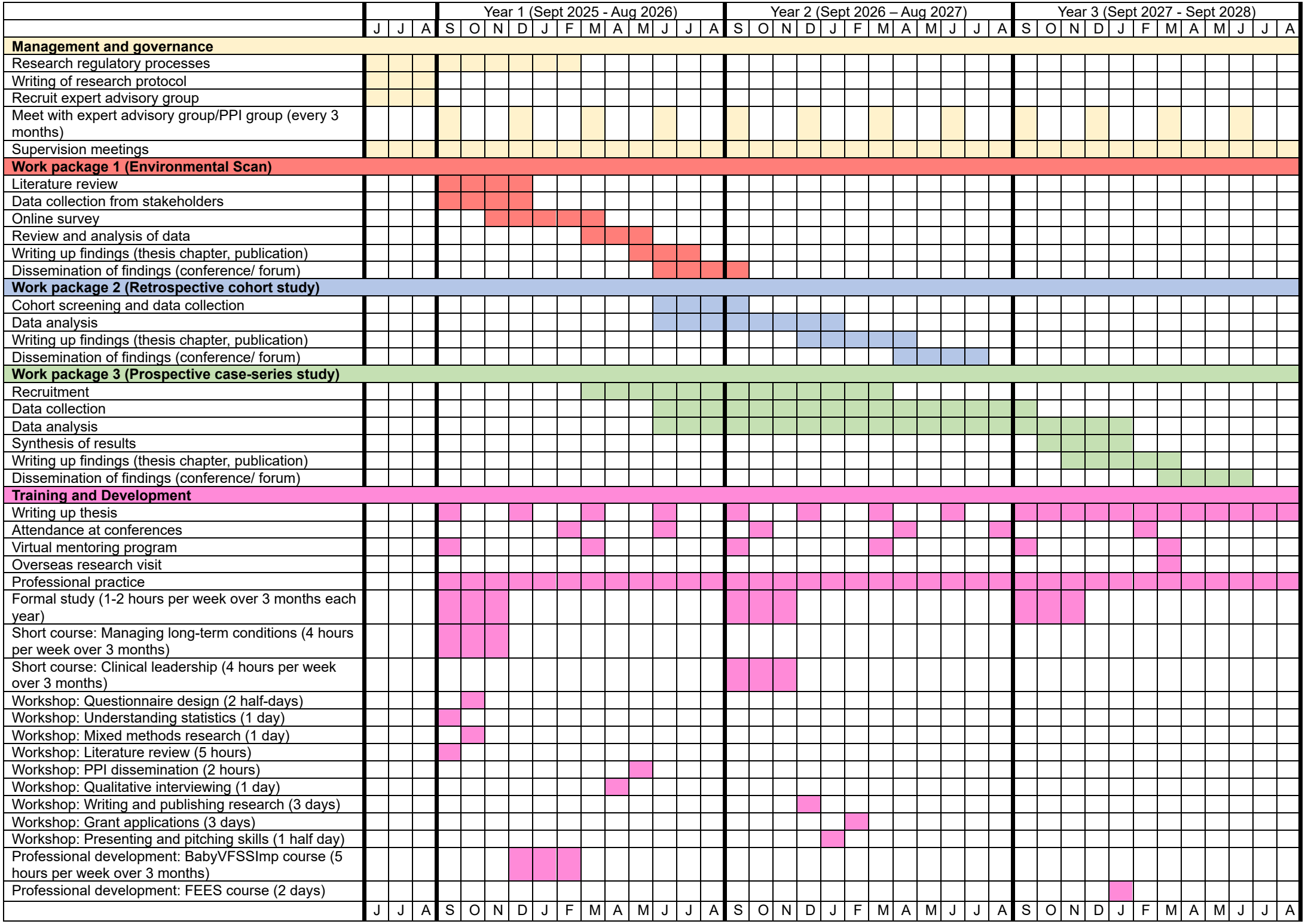
# Appendix #2: Gantt Chart SL.pdf


Year 1 (Sept 2025 - Aug 2026)
Year 2 (Sept 2026 – Aug 2027)
Year 3 (Sept 2027 - Sept 2028)
J J A S O N D J F M A M J J A S O N D J F M A M J J A S O N D J F M A M J J A
Management and governance
Research regulatory processes
Writing of research protocol
Recruit expert advisory group
Meet with expert advisory group/PPI group (every 3 months)
Supervision meetings
Work package 1 (Environmental Scan)
Literature review
Data collection from stakeholders
Online survey
Review and analysis of data
Writing up findings (thesis chapter, publication)
Dissemination of findings (conference/ forum)
Work package 2 (Retrospective cohort study)
Cohort screening and data collection
Data analysis
Writing up findings (thesis chapter, publication)
Dissemination of findings (conference/ forum)
Work package 3 (Prospective case-series study)
Recruitment
Data collection
Data analysis
Synthesis of results
Writing up findings (thesis chapter, publication)
Dissemination of findings (conference/ forum)
Training and Development
Writing up thesis
Attendance at conferences
Virtual mentoring program
Overseas research visit
Professional practice
Formal study (1-2 hours per week over 3 months each year)
Short course: Managing long-term conditions (4 hours per week over 3 months)
Short course: Clinical leadership (4 hours per week over 3 months)
Workshop: Questionnaire design (2 half-days)
Workshop: Understanding statistics (1 day)
Workshop: Mixed methods research (1 day)
Workshop: Literature review (5 hours)
Workshop: PPI dissemination (2 hours)
Workshop: Qualitative interviewing (1 day)
Workshop: Writing and publishing research (3 days)
Workshop: Grant applications (3 days)
Workshop: Presenting and pitching skills (1 half day)
Professional development: BabyVFSSImp course (5 hours per week over 3 months)
Professional development: FEES course (2 days)
J J A S O N D J F M A M J J A S O N D J F M A M J J A S O N D J F M A M J J A

# Appendix #3: SoECAT Tool A Sabrena Lee_authorised_EF_4.6.2024.xlsx